

Firoozbakht’s Conjecture: A Structured Attack

A machine-checked reduction and a machine-checked barrier, an unconditional finite-range architecture, the closure of the Riemann-Hypothesis route, a refuted pruning lemma, and what remains open

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Abstract

Firoozbakht’s conjecture asserts that $p_{n+1}^{1/(n+1)} < p_n^{1/n}$ for every $n \geq 1$, i.e. that $n \mapsto p_n^{1/n}$ is strictly decreasing. It is open. This paper reports a two-round structured attack that neither proved nor refuted it, and states precisely what the attack did establish. Five things. (i) The reduction of the conjecture to the prime-gap inequality $g_n < T_n$, where $T_n := p_n(p_n^{1/n} - 1)$, is *machine-checked* in Lean 4 against Mathlib, and so is a *barrier*: Bertrand’s postulate — the only prime-gap bound Mathlib carries — is proved insufficient at *every* index $n \geq 2$, the ceiling sitting strictly above the threshold it would have to fall below. The development builds with exactly one `sorryAx` dependent over 63 declarations, namely the conjecture itself. (ii) The pruning lemma on which record-gap verification searches rest was carrying three inequivalent statements under one name. Named apart, the strongest is *false*, with two exhibited witnesses and a 17-index exception census below 10^9 ; the two survivors are proved *formally incomparable*; and the pruning is undamaged, because a first-failure-maximality theorem shows it consumes either survivor at a single index. (iii) An unconditional finite-range theorem, resting on one explicit estimate for $\pi(x)$ due to Dusart, converts any first-occurrence prime-gap table into a proof of the conjecture up to that table’s reach; it reproduces the published constant 1920 at 2^{64} with no tuning, and comes with a table-free window $396\,738 \leq p_n \leq 777\,600$ and a proof that the window closes permanently. Its companion near-record theorem is repaired and tightened: first-failure maximality now holds against all primes below $0.94970\,p_{n_0}$ on Dusart alone and below $0.998244\,p_{n_0}$ on Axler’s corollaries, which this run has now read at the locator in both editions. (iv) Five theorems close the Riemann-Hypothesis route *as a route*, with the quantifier that the first round dropped restored: the sharpest published RH-conditional gap bound certifies the Firoozbakht inequality at exactly one index of the range where the bound is available, no envelope of power type at any positive exponent certifies it beyond finitely many indices, the critical constant on the $\sqrt{p} \log p$ scale is exactly $2/e$, and Cramér’s lim sup hypothesis does not entail the conjecture over integer sequences. (v) An exhaustive sweep to 10^{11} (4 118 054 812 consecutive prime pairs) finds no violation and no near-miss, with $\max_{n \geq 10} g_n/T_n = 0.8318$; two further independent sweeps in round 2 reproduce it. We also state, at length, what the attack did *not* establish: the conjecture is open, the underlying corpus’s evidence gate stands at **BLOCKED** on three named unreconciled seams, and no citation clearance exists for this paper.

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1 Introduction

1.1 The conjecture

Let $p_1 = 2 < p_2 = 3 < p_3 = 5 < \dots$ enumerate the primes, one-indexed.

Definition 1.1 (Firoozbakht’s conjecture, F). F is the assertion

$$p_{n+1}^{1/(n+1)} < p_n^{1/n} \quad \text{for every } n \geq 1, \quad (1)$$

equivalently: the sequence $(p_n^{1/n})_{n \geq 1}$ is strictly decreasing.

The conjecture is due to Farideh Firoozbakht (University of Isfahan, 1982) and was never published by her [Fir82]; its first appearance in print is [Rib04, p. 185]. Its date and unpublished status are attested independently by [Kou15a, §1], [FM18, §1] and [Vis19]. Sun states it in the form “the sequence $(\sqrt[n]{p_n})_{n \geq 1}$ is strictly decreasing” [Sun13, §1]; Kourbatov states the equivalent $p_{k+1} < p_k^{1+1/k}$ [Kou15a, §1, eq. (1)]. All of these are the same statement, and we use them interchangeably.

1.2 Why it is interesting, and why it is hard

F is a statement about prime gaps in disguise. Writing $g_n := p_{n+1} - p_n$ and $L_n := \log p_n$ (natural logarithm throughout), F is exactly the assertion that g_n stays below the threshold $T_n := p_n(p_n^{1/n} - 1)$ at every index (Lemma 2.1), and $T_n = L_n^2 - L_n - 1 + o(1)$ [Kou15a, §5, Thm. 5]. So F implies a Cramér-scale gap bound with the leading constant 1:

$$F \implies (g_k < L_k^2 - L_k - 1 \text{ for all } k > 9), \quad (2)$$

which is [Kou15a, §2, Thm. 1]; see also [FM18, Thm. 2.2] and the weaker +1 variant in [Sun13, §1]. In the converse direction,

$$(g_k < L_k^2 - L_k - 1.17 \text{ for all } k > 9, p_k \geq 29) \implies F, \quad (3)$$

which is [Kou15a, §4, Thm. 3]. Thus F is trapped between two $\log^2$ -scale gap bounds differing by an additive 0.17.

That sandwich is the source of the difficulty. The best *unconditional* prime-gap bound available is $g_n = O(p_n^{0.525})$ [BHP01]; under the Riemann Hypothesis the state of the art is $g_n \leq \frac{22}{25} \sqrt{p_n} \log p_n$ for $p_n > 3$ [CMS19, §1.2, Thm. 5]. Both are *powers* of p_n ; F needs a *polylogarithmic* bound. The distance is square-root scale against log scale, and no known

method bridges it, even conditionally on RH. Section 7 turns this observation into theorems, and §3 turns the formal half of it into a kernel-checked one.

On the other side, the standard heuristic points the other way. Cramér’s random model, in the corrected form Granville gives it, suggests

$$\max_{p_n \leq x} (p_{n+1} - p_n) \gtrsim 2e^{-\gamma} \log^2 x, \quad 2e^{-\gamma} = 1.12292 \dots,$$

[Gra95, preprint p. 12, after eq. (20)], which is *incompatible* with F via (2). At least one of the two — Firoozbakht’s conjecture or the Cramér–Granville heuristic — must fail, and no present technique says which. This is the single most important framing sentence of the subject, and we return to it in §10.

1.3 Literature

The relevant literature falls into five groups.

Statement, strengthenings and consequences. Ribenboim records the conjecture [Rib04, p. 185]; Sun uses it as the motivating contrast for a provable analogue about sums of primes [Sun13, Thm. 2.1]. Ferreira and Mariano derive a systematic list of consequences [FM18]: the necessary gap bound ([FM18, Thm. 2.2]), $g_n < \sqrt{n}$ for $n \geq 3645$ ([FM18, Lem. 3.2]), a Sierpiński-type consequence about primes in rows of the $n \times n$ array of $1, \dots, n^2$ ([FM18, Cons. 3.3]), and — most importantly for us — the only unconditional positive statement in the literature about F itself, Theorem 2.4 below. Visser [Vis19] states and verifies the two known strengthenings, Nicholson’s (traced to an OEIS comment [Slob], otherwise unpublished) and Farhadian’s [FJ17], and records the implication chain Farhadian $\Rightarrow$ Nicholson $\Rightarrow$ Firoozbakht [Vis19, Conj. 2], proved in [FM18, Thm. 4.5] together with Forgues’ weaker variant.

Criteria. The necessary and sufficient conditions (2) and (3) are Kourbatov’s [Kou15a, §2 Thm. 1, §4 Thm. 3], together with a family of sufficient conditions with $b \rightarrow 1$ [Kou15a, §4, Thm. 4] that presuppose the finite verification (they are stated for $p_k > 4 \cdot 10^{18}$). Both rest on Axler’s effective bounds for $\pi(x)$ [Axl14]. That dependency was the largest unopened source of the first round; it is now opened, read at the locator in both editions, and carries a fresh and purely editorial hazard instead (Caveat 5.5).

Verification. Kourbatov verified F for all $p_k < 4 \cdot 10^{18}$ [Kou15b, §4, Thm.], using the first-occurrence prime-gap table of Oliveira e Silva, Herzog and Pardi [OHP14]; the 2023 endnotes to [Kou15b] extend the range to $2^{64} \approx 1.8447 \cdot 10^{19}$ and record that “prime gaps of size $g < 1920$ cannot violate” the conjecture. Visser independently reports verification up to the 81st maximal prime gap, “certainly for all primes $p < 2^{64}$ ” [Vis19, Abstract, §1]. We reconstruct the architecture of this verification from first principles in §6.

The heuristic side. Cramér proves $\limsup (P_{n+1} - P_n) / (\log P_n)^2 = 1$ with probability 1 for his urn model and only *suggests* the analogue for the primes [Cra36, pp. 24, 26–27]; this distinction is not a pedantry and we preserve it everywhere. Granville corrects the model and obtains the opposite prediction [Gra95, preprint pp. 10, 12]. The empirical record of the Cramér–Shanks–Granville ratio $(q - p) / \log^2 p$ is A111943 [Sloa]; Shanks conjectured the ratio never reaches 1 [Sha64], Granville the opposite. The record stands at 0.9206 at $p = 1\,693\,182\,318\,746\,371$ (gap 1132, B. Nyman, 1999) [Sloa].

Large gaps. The best lower bound on the largest gap below X is

$$G(X) \geq f(X) \frac{\log X \log \log X \log \log \log \log X}{(\log \log \log X)^2}, \quad f(X) \rightarrow \infty,$$

[For+16] — an iterated-log factor above $\log X$, but still a full power of $\log$ below the $\log^2 X$ a counterexample to F would need.

1.4 What changed between the two rounds of this attack

This paper is the round-2 state of the work and supersedes, in place, a round-1 paper of the same name dated 25 July 2026. The change is not cosmetic and the honest summary is mixed.

What changed. Round 1 closed with two named blocking defects in its own corpus. Both are now *mathematically repaired*, by independent re-derivation rather than by re-assertion, and every constant below was recomputed from the statements by a reviewing leg that shared no code path with the leg being reviewed.

1. The first defect was a vocabulary collision: the pruning lemma that record-gap searches consume was carrying three inequivalent statements under one name. Naming them apart (§4) did more than disambiguate — it showed the strongest of them to be **false** (Theorem 4.3), showed the two survivors to be **incomparable** rather than ordered (Proposition 4.8), and exposed a fourth statement nobody had been measuring under its own name (Definition 4.1). Round 1’s paper asserted a strict ordering among these statements; that assertion was wrong and is withdrawn here.
2. The second defect was a mis-derived upper bound on T_m , false by a factor ≈ 38.8 over part of its range, which round 1 declined to state a constant for. It is now restated in its tight form and *proved* rather than checked numerically, in two independent ways (§6.5). Round 1’s suppressed constant is superseded twice over, and the sharper of the two repairs is carried here.

Three further results are new: the machine-checked Bertrand barrier (§3), the first-failure-maximality theorem that makes the refutation of the pruning lemma harmless (Theorem 4.11), and the restored quantifier in the RH route, which dissolved a silent contradiction between two artifacts of the run (§7.2).

What did not change. F is open — neither proved nor refuted by either round. The Lean **sorry** count is still exactly one, and it is the conjecture. The obstruction analysis of §11 is unchanged and was *reinforced*, in three independent places, by round 2. The underlying corpus’s evidence gate is still **BLOCKED**; the reason changed, from unrepaired mathematical defects to unreconciled repairs, and §11.3 argues that this is a worse signal rather than a better one. No citation clearance exists for this paper, in either round.

1.5 What this paper claims, and the vocabulary of its claims

This paper is an honest report on an attack that did not settle the conjecture. To keep the strength of each statement legible we fix the following vocabulary and use it without exception.

Code	Meaning
[K]	Machine-checked by the Lean 4 kernel against Mathlib (§3).
[P]	Paper proof, derived in this work, independently re-derived by an adversarial review leg.
[P·s]	As [P], but resting somewhere on a source that was not opened in this work, or whose ledger status is contested (§11.4).
[C]	Finite computation, exhaustive, independently reproduced by two or more code paths.
[H]	Heuristic. Not evidence for or against anything, and never used in a proof.

The word *proved*, unqualified, is reserved in this paper for statements carrying [K]: that is Lemma 2.1 and the three barrier theorems of §3, and nothing else. Everything at [P] or [P·s] is a paper proof and is labelled as such at the point of use. In particular:

F is neither proved nor refuted in this paper, and nothing below should be read as evidence that either is close. The defensible summary sentence is: *Firoozbakht’s conjecture is numerically robust over the verified range and simultaneously incompatible with the standard Cramér–Granville heuristic; at least one of the two must fail, and no current technique can say which.*

1.6 Contributions

1. [K] A Lean 4/Mathlib development in which the reduction $F \leftrightarrow (\forall n \geq 1, g_n < T_n)$ is machine-checked, with an exhaustive list-free axiom audit over 63 declarations finding exactly one `sorryAx` dependent — the conjecture itself — together with three `sorry`-free theorems establishing that the only prime-gap input Mathlib carries is insufficient at every index $n \geq 2$ (§3).
2. [P] The refutation of the pruning lemma the attack had ranked as its most tractable open obligation, the separation of the two survivors, and a first-failure-maximality theorem showing the pruning route consumes either survivor at a single index and is therefore undamaged (§4).
3. [P] The monotone-bar principle and its truncated form, which discharge the practical content of that obligation against a monotone surrogate whose oscillation is nil (§5).
4. [P], [C] An unconditional finite-range theorem (Theorem 6.4) resting on a single explicit $\pi(x)$ estimate, reproducing the published constant 1920 at 2^{64} ; and a table-free window with a proof that it closes permanently (§6).
5. [P·s] A repaired and tightened near-record maximality theorem in two branches (Theorems 6.14 and 6.15), replacing a round-1 bound that did not follow from its stated justification (§6.5).
6. [P] Five theorems closing the RH route as a route (§7), including a Lambert- W determination of the critical constant $2/e$, a counter-model showing that Cramér’s limsup hypothesis does not entail F over integer sequences, and the per-index critical constant that reconciles two artifacts of the run.
7. [C] An exhaustive verification to 10^{11} , reproduced by two further independent round-2 sweeps, with a discipline that *raises* rather than returns when a margin lands inside its error budget (§9).

8. An explicit account of what is *not* established, including three named unreconciled seams in the underlying corpus, two live labelling defects inside the repairs, and the absent citation audit (§11).

2 Setup

2.1 Notation

Throughout, p_n is the n -th prime with $p_1 = 2$ (one-indexed), $\pi(x)$ is the prime-counting function, so that

$$\pi(p_n) = n, \quad (4)$$

and

$$L_n := \log p_n, \quad g_n := p_{n+1} - p_n, \quad T_n := p_n(p_n^{1/n} - 1) = p_n(e^{L_n/n} - 1).$$

All logarithms are natural. We write $\rho_n := g_n/T_n$ for the *normalised objective*; F at n is exactly $\rho_n < 1$. We call n a *failure* if $g_n \geq T_n$, and a *record index* (equivalently a maximal-gap index) if $g_n > g_j$ for every $j < n$.

Identity (4) looks like bookkeeping and is not: it is the hinge on which every analytic estimate in this paper turns, because T_n depends on the pair (p_n, n) and the only way to control n analytically is through π .

2.2 The reduction

Lemma 2.1 (Gap form of F ; [K]). *For every $n \geq 1$,*

$$p_{n+1}^{1/(n+1)} < p_n^{1/n} \iff p_{n+1}^n < p_n^{n+1} \iff g_n < T_n.$$

Proof. Idea: Raise to the power $n(n+1)$, take logarithms, and subtract p_n from both sides; every step is an equivalence because the maps involved are strictly increasing.

All quantities are real and exceed 1. The map $t \mapsto t^{n(n+1)}$ is strictly increasing on $(0, \infty)$, so raising $p_{n+1}^{1/(n+1)} < p_n^{1/n}$ to that power gives $p_{n+1}^n < p_n^{n+1}$, and the step is reversible. Applying the strictly increasing $\log$ gives $n \log p_{n+1} < (n+1) \log p_n$, i.e. $\log p_{n+1} < L_n(1 + 1/n)$, i.e. $p_{n+1} < p_n e^{L_n/n}$, i.e. $p_n + g_n < p_n + p_n(e^{L_n/n} - 1)$. Subtracting p_n gives $g_n < T_n$. $\square$

Lemma 2.1 is formalised and machine-checked in Lean 4 (§3), together with the intermediate equivalence, and both equivalences are pinned to numerals ($g_1 = 1$, $T_1 = 2^2 - 2$) so that a mis-stated g or T fails to compile rather than passing quietly.

Remark 2.2 (Strict versus non-strict). Visser states the conjecture with $\leq$ [Vis19, Conj. 3, eq. (2.4)]. The two forms coincide if and only if T_n is never an integer. That is true for $n \geq 2$: $p_n^{1/n}$ rational would force it to be an integer and p_n to be a perfect n -th power, which unique factorisation forbids for a prime. We nonetheless use the strict form throughout and never silently convert between them.

2.3 The two known unconditional facts about F

Theorem 2.3 (Verified range; [C], cited). *F holds for every n with $p_n < 2^{64} = 18\,446\,744\,073\,709\,551\,616 \approx 1.8447 \cdot 10^{19}$.*

This is [Kou15b, endnotes, 5 Jan 2023], corroborated by [Vis19, Abstract, §1 eq. (1.4)]. Its computational input is the first-occurrence gap table of [OHP14]. *This paper does not re-establish Theorem 2.3*; §6 reconstructs the implication from a table to the range, and §11.4 records that the table itself was never opened in this work. Every statement below that

consumes $p_{n_0} > 2^{64}$ consumes this theorem, and therefore that unopened table; the places where it happens are named at the point of use rather than collected here.

Theorem 2.4 (Infinitely often; cited). *There are infinitely many $n \in \mathbb{N}$ with $p_n^{1/n} > p_{n+1}^{1/(n+1)}$.*

This is [FM18, Thm. 5.2], proved unconditionally there via Zhang’s bounded-gaps theorem (cited as [FM18, Thm. 5.1]). It is the only unconditional theorem in the literature that says something positive about F itself.

Caveat 2.5. Theorem 2.4 says *infinitely often*, not *eventually*. It is fully compatible with infinitely many failures. It must never drift into “ F holds for large n ”.

3 The formal substrate, and a machine-checked barrier

3.1 What is machine-checked

The reduction chain

$$\text{Conjecture} \leftrightarrow \text{ConjectureReal} \leftrightarrow (\forall n \geq 1, g_n < T_n)$$

— i.e. Lemma 2.1 — is formalised in Lean 4 against Mathlib and checked by the kernel. Toolchain: leanprover/lean4:v4.29.0; Mathlib tag v4.29.0, revision 8a178386ffc0f5fef0b77738bb5449d50efeea95.

Gate	Result
lake build exit code	0 (2208 jobs)
build warnings	1 — the declared open target, and nothing else
declarations scanned by the axiom audit	63 (exhaustive, list-free; was 60 before the barrier file)
declarations depending on <code>sorryAx</code>	1 — <code>Firoozbakht.firoozbakht</code>
live <code>sorry</code> tokens in sources	1
stray <code>axiom</code> , <code>native_decide</code> , @[implemented_by] or <code>unsafe</code>	none (grep-clean)
<code>Statement.lean</code> SHA-256, before and after	identical (6528868823c0637d...)

The single `sorryAx` dependent is Firoozbakht’s conjecture itself, which round 2 attempted and failed to prove; `exact?`, `aesop` and `decide` all fail, as expected. *An attack on an open Π_1 statement whose `sorry` count reaches zero would be reporting a fabrication, not a result.* The byte-identity of the statement file across the round is what forecloses the failure mode “closed a `sorry` by weakening the statement”; the equivalences were also shown non-vacuous, $g_1 = 1$ and $T_1 = 2^2 - 2$ being pinned to numerals.

Two things were tested rather than trusted. First, the audit is list-free: a hand-maintained list of declarations to `#print axioms` is a hazard dressed as a check, since a `sorry` in a forgotten declaration is invisible. Second, the *detector* was tested by planting a `sorry` in the enlarged tree; the audit then scanned 64 declarations and reported two names, after which the plant was deleted and the tree returned to 63 and one. So the one-entry unproved list is produced by a detector demonstrated to fire. Separately, the checker was tested against a corpus of 27 adversarial statements, all false or ill-formed; all 27 behaved as specified, and the one corpus entry caught by *no* gate in this study — a true-but-differently-meant statement produced by an $\mathbb{N} \rightarrow \mathbb{R}$ coercion — is named rather than omitted.

3.2 The Bertrand barrier

Mathlib’s strongest prime-gap result is Bertrand’s postulate. Round 2 asked what that buys, and proved that the answer is: nothing, at every index but the first.

Theorem 3.1 (The Bertrand barrier; [K]). *The following are theorems of the development, all `sorry-free`.*

- (a) `bertrand_gap`: $p_{n+1} \leq 2p_n$ for $n \geq 1$.
- (b) `p_lt_two_pow`: $p_n < 2^n$ for $n \geq 2$.
- (c) `bertrand_ceiling_above_threshold`: $p_n^{1+1/n} < 2p_n$ for every $n \geq 2$.

Read (c) in the right direction. For Bertrand to *certify* F at n — i.e. for the chain $p_{n+1} \leq 2p_n < p_n^{1+1/n}$ to close — its ceiling $2p_n$ would have to sit *below* the threshold $p_n^{1+1/n}$. It sits strictly *above* it, at every $n \geq 2$. The mechanism in one line: Bertrand supplies constant multiplicative slack 2, while the threshold’s slack $p_n^{1/n} = \exp(L_n/n)$ tends to 1; certification would need $2^n \leq p_n$, and $p_n < 2^n$, which is (b) — itself proved *from* Bertrand. **The best available tool proves its own insufficiency.**

Remark 3.2 (What the barrier is, and what it is not). This is a negative-capability result. It says nothing about whether F is true, and nothing about whether F is provable; it says that one specific route — the only prime-gap input Mathlib currently carries — cannot reach it, and it says so at every index rather than “for large n ”. It also says nothing about routes through bounds Mathlib does *not* have: $p^{0.525}$ of [BHP01], the RH-conditional $\sqrt{p} \log p$ of [CMS19], or a Cramér-scale $\log^2 p$. Those are examined on paper in §7, where the first two are found insufficient as well and the third is found not to exist in the literature. Round 1 recorded “Bertrand is useless here” as an observation; the difference round 2 makes is that there is now no sharper version of Bertrand worth formalising, because the shortfall is a whole scale and not a constant.

3.3 What is not formalised, and why

- **The smooth model** (Proposition 8.1). Fully within Mathlib’s reach; still not done after two rounds, and still the highest-leverage formalisation target the work has.
- **The limsup corollary**. Needs effective $\pi(x)$ bounds that are not assumed present in Mathlib.
- **Any verified range past $n \leq 4$** . This is a hard limit and deserves naming. `Nat.nth` is *noncomputable* [Thea] — not merely slow — so `decide` cannot produce p_n at all; and Mathlib’s prime-specific `nth` API is exactly five `@[simp]` base lemmas, `nth Prime 0 = 2` through `nth Prime 4 = 11` [Theb]. Extending requires building `Nat.count ↔ Nat.nth` bridging machinery, plus prime literals with `Nat.Prime` certificates and a “no prime strictly between” lemma for each consecutive pair. Reporting a larger verified N without that work would be a fabrication.
- **Dusart’s estimate** (Lemma 6.1) and Axler’s corollaries are not in Mathlib and would have to enter any formalisation of §6 as explicit hypotheses — and be declared as such in any paper quoting the result.

Caveat 3.3 (The missing input, named precisely). Discharging the one remaining `sorry` needs a gap bound of Cramér strength, $g_n < p_n^{1+1/n} - p_n \approx L_n^2$. No such unconditional theorem exists in the literature. **Even a complete formalisation of every published prime-gap**

bound, unconditional or RH-conditional, would not discharge it: the obstruction is mathematical first and formalisation-budget second. This is the one place where a reader might expect “more Lean would help”, and it would not.

Caveat 3.4 (Indexing). Mathlib’s `nth` is 0-indexed (`nth Prime 0 = 2`) while the conjecture as posed here is 1-indexed ($p_1 = 2$). Every index threshold in the literature — $k > 9$, $n \geq 10$, $n \geq 3645$, $n \geq 5$ — is stated in the 1-indexed convention. The off-by-one must be fixed once, in the statement file; it is the single most likely source of a silent error in a formalisation of this material.

4 First-failure maximality: four predicates, one of them false

4.1 The obligation, and the collision

Every practical verification of F over a range checks *record* gaps rather than all indices. What licenses that pruning is

(M1) — first-failure maximality. If F is false, its least failure n_0 is a *strict* record index: $g_j < g_{n_0}$ for every $j < n_0$.

Round 1 ranked the statement usually offered in support of (M1) as its most tractable open obligation, under the name (P6’), and reported it as “open, and empirically unviolated by every measurement that bears on it”. Both halves of that sentence were wrong, and for the same reason: the name carried three inequivalent statements, so no single measurement bore on the prose. We name them apart, once and for all.

Definition 4.1 (The four predicates). For $n \geq 1$ put

$$r(n) := \max\{j \leq n : j \text{ is a record index}\}, \quad \mu(n) := \min\{j : g_j \geq g_n\}.$$

Both are well defined ($j = 1$ is a record; $j = n$ dominates n). The four statements are:

- (P6’-pair): $T_m \leq T_n$ whenever $m < n$ and some record index j satisfies $m \leq j < n$;
- (P6’-gov): $T_{r(n)} \leq T_n$ for every n ;
- (P6’-min): $T_{\mu(n)} \leq T_n$ for every n ;
- (P6’-rec): $T_j \leq T_{j'}$ for consecutive record indices $j < j'$.

In (P6’-pair) the index m is *arbitrary* — that is exactly what “ p_m and p_n straddle a record gap” says, and it is the reading round 1’s prose carried. (P6’-gov) is vacuous at record indices, so it is a statement about the indices strictly *after* a record. (P6’-rec) is the fourth statement, which round 1 never isolated and which turns out to be the hinge of the lattice.

Lemma 4.2 ([P]). $\mu(n)$ is a record index, and $\mu(n) \leq r(n) \leq n$.

Proof. Idea: Minimality of $\mu(n)$ forbids an earlier index with a gap as large; and the running maximum at n is attained at $r(n)$, which therefore competes in the minimisation.

If $\mu(n)$ were not a record there would be $j < \mu(n)$ with $g_j \geq g_{\mu(n)} \geq g_n$, contradicting minimality. For the second part, $g_{r(n)} \geq g_n$ because $r(n)$ carries the running maximum of g on $[1, n]$, so $r(n)$ belongs to the set $\mu(n)$ minimises over. $\square$

4.2 The strongest predicate is false

Theorem 4.3 ((P6'-pair) is false; [C]). *There exist indices $m < j < n$ with j a record index and $T_m > T_n$. Hence (P6'-pair) is false, and it is false already on $[1, 1847]$.*

Proof. Idea: Exhibit a pair straddling a record gap in the strictest available sense, and compute both bars.

Two witnesses, both computed at 60 decimal digits from the definition $T_k = p_k(e^{L_k/k} - 1)$ with p_k read off an independent sieve.

	role	index	prime	value
W1	m	1823	15 641	$T_m = 83.0807167192698\dots$
	j (record)	1831	15 683	$g_j = 44$ (12th maximal gap)
	n	1847	15 823	$T_n = 83.0521061144139\dots$
W2	m	10 655 449	191 912 639	$T_m = 343.5112716866658\dots$
	j (record)	10 655 462	191 912 783	$g_j = 248$ (27th maximal gap)
	n	10 655 590	191 915 033	$T_n = 343.5112358945690\dots$

For W1, $p_m = 15\,641 < 15\,683 = p_j < p_{j+1} = 15\,727 < 15\,823 = p_n$, so the whole gap interval (p_j, p_{j+1}) lies between p_m and p_n : the pair straddles the record gap in the strictest available sense. And

$$T_m - T_n = +0.02861060485582\dots > 0, \quad \text{respectively} \quad T_m - T_n = +3.5792097 \cdot 10^{-5} > 0.$$

Either witness refutes (P6'-pair). □

Remark 4.4 (The witnesses are not floating-point). W1's margin is $2.9 \cdot 10^{-2}$ against a bar $T \approx 83$, whose `float64` unit in the last place is $1.4 \cdot 10^{-14}$: the margin is $2 \cdot 10^{12}$ ulps. W2's is $6.3 \cdot 10^8$ ulps. Both were nevertheless recomputed at 60 digits and independently reproduced by a reviewing leg, because the escalation discipline of §9 applies here *with the sign flipped*: a spurious exception would be the reassuring outcome for the attack, and the leg reporting it is the leg that wanted it to be false.

Proposition 4.5 (Exception census; [C]). *Below 10^9 there are exactly 17 indices n admitting an m that witnesses a failure of (P6'-pair), in exactly two clusters: $n \in \{1836, 1837, 1840, 1844, 1845, 1846, 1847\}$, and a second cluster of ten indices in $[10\,655\,562, 10\,655\,594]$. Both clusters sit a few indices after a maximal gap. The extreme margin over all admissible pairs is $-2.861 \cdot 10^{-2}$, attained at W1.*

Caveat 4.6 (A labelling defect in the census, carried rather than smoothed). The census counts *indices* n , not *pairs* (m, n) : for a fixed n there may be several witnessing m . The upstream artifact reporting it calls it a “complete census of admissible pairs”, which it is not; as a list of n it is complete. The defect is one of labelling, not of arithmetic, and it does not touch Theorem 4.3, which needs a single witness. It is recorded in §11.3 as one of the corpus's live defects.

4.3 The lattice, and the invalid link

Proposition 4.7 (Valid implications; [P]). $(\text{P6'-pair}) \Rightarrow (\text{P6'-gov}); \quad (\text{P6'-pair}) \Rightarrow (\text{P6'-rec});$
and $(\text{P6'-gov}) \wedge (\text{P6'-rec}) \Rightarrow (\text{P6'-min})$.

Proof. Idea: The first two are instances of (P6'-pair) with m chosen to be a record index; the third chains (P6'-rec) along the record indices from $\mu(n)$ up to $r(n)$ and then applies (P6'-gov) once.

For (P6'-gov): fix n . If $r(n) = n$ the statement is $T_n \leq T_n$. Otherwise set $m := j := r(n)$; then $m < n$ and $m \leq j < n$, so the pair is admissible and (P6'-pair) gives $T_{r(n)} \leq T_n$. For (P6'-rec): let $j < j'$ be consecutive records and take $m := j$, $n := j'$; the pair is admissible. For the third: by Lemma 4.2, $\mu(n)$ is a record index and $\mu(n) \leq r(n)$; both are records, so iterating (P6'-rec) along the record indices from $\mu(n)$ to $r(n)$ gives $T_{\mu(n)} \leq T_{r(n)}$, and (P6'-gov) at n gives $T_{r(n)} \leq T_n$. Chain. $\square$

Proposition 4.8 (The two survivors are incomparable; [P]). *In the setting in which all four predicates are stated — a strictly increasing integer sequence with gap sequence (g_k) and an arbitrary real bar (T_k) — both $(\text{P6'-gov}) \Rightarrow (\text{P6'-min})$ and $(\text{P6'-min}) \Rightarrow (\text{P6'-gov})$ are **false**.*

Proof. Idea: Four indices suffice: choose gaps so that the governing record and the minimal dominating index differ at the last index, then choose the bar to break whichever implication is being tested.

Take $g = (2, 4, 6, 3)$. The record indices are 1, 2, 3; $r(4) = 3$ while $\mu(4) = 2$ (the least j with $g_j \geq 3$), so index 4 separates the two predicates.

- $(\text{P6'-gov}) \not\Rightarrow (\text{P6'-min})$: take $T = (0, 10, 1, 5)$. (P6'-gov) is vacuous at $n = 1, 2, 3$ (record indices) and at $n = 4$ reads $T_3 = 1 \leq 5 = T_4$, true. (P6'-min) at $n = 4$ reads $T_2 = 10 \leq 5 = T_4$, false.
- $(\text{P6'-min}) \not\Rightarrow (\text{P6'-gov})$: take $T = (0, 1, 10, 5)$. (P6'-min) is trivial at $n = 1, 2, 3$ (there $\mu(n) = n$) and at $n = 4$ reads $T_2 = 1 \leq 5 = T_4$, true. (P6'-gov) at $n = 4$ reads $T_3 = 10 \leq 5 = T_4$, false.

$\square$

Two consequences, both of which correct round 1.

Remark 4.9 (The withdrawn ordering). Round 1's paper and its fault report both asserted that the three statements are strictly ordered, $(\text{P6'-pair}) \Rightarrow (\text{P6'-gov}) \Rightarrow (\text{P6'-min})$, and that the pruning consumes the weakest. **The second link is invalid** (Proposition 4.8); the missing ingredient is (P6'-rec), a fourth statement nobody had been measuring under its own name. The corrected picture is

$$(\text{P6'-pair}) \implies (\text{P6'-gov}) \wedge (\text{P6'-rec}) \implies (\text{P6'-min}),$$

with (P6'-pair) **false** and (P6'-gov), (P6'-rec), (P6'-min) all open. The reason the invalid link is not pedantry: (P6'-gov) is vacuous at every record index, so it says nothing whatsoever about how T moves from one record to the next — which is exactly the step (P6'-min) needs. Any proof of $(\text{P6'-gov}) \Rightarrow (\text{P6'-min})$ must therefore consume a property of the primes. This is the same species of finding as the counter-model of §9.6: the separation is *structural*, visible with no arithmetic input.

Remark 4.10 (And it is not the case that (P6'-min) is “the weakest”). Upstream prose in the corpus calls (P6'-min) “the weakest of the three” and “the easier obligation”, on the strength of the ordering Proposition 4.8 disproves. The correct statement is: (P6'-min) is the one Theorem 4.11 needs and the one whose empirical margin does not decay, it is *incomparable* to (P6'-gov), and (P6'-rec) must be listed beside it because Proposition 4.7 needs it. The upstream prose has not been amended; see §11.3.

4.4 Why the refutation costs nothing

Theorem 4.11 (First-failure maximality from either survivor, at one index; [P]). *Suppose F is false and let n_0 be its least failure. If either $T_{r(n_0)} \leq T_{n_0}$ or $T_{\mu(n_0)} \leq T_{n_0}$, then $g_j < g_{n_0}$ for every $j < n_0$ — that is, (M1) holds.*

Proof. Idea: Both branches are three-link chains: minimality of n_0 turns “earlier” into “below its own bar”, the predicate transports the bar forward, and being a failure closes the loop into $g_{n_0} < g_{n_0}$.

The (P6'-min) branch. Put $m := \mu(n_0)$, so $g_m \geq g_{n_0}$ by definition. Suppose $m < n_0$. By minimality of n_0 , m is not a failure, so $g_m < T_m$. Then

$$g_{n_0} \leq g_m < T_m = T_{\mu(n_0)} \leq T_{n_0} \leq g_{n_0},$$

the last step because n_0 is a failure — a contradiction. Hence $m = n_0$, i.e. $\min\{j : g_j \geq g_{n_0}\} = n_0$, i.e. $g_j < g_{n_0}$ for every $j < n_0$.

The (P6'-gov) branch. If $r(n_0) = n_0$ then n_0 is a record index and the conclusion is the definition. Otherwise $r(n_0) < n_0$, so by minimality $g_{r(n_0)} < T_{r(n_0)} \leq T_{n_0} \leq g_{n_0}$; and for any $j < n_0$ we have $g_j \leq g_{r(n_0)}$ because $r(n_0)$ carries the running maximum of g on $[1, n_0]$. Chaining gives $g_j \leq g_{r(n_0)} < g_{n_0}$. $\square$

Three consequences, and they are why Theorem 4.3 is a clarification rather than a loss.

1. **The pruning never needed (P6'-pair).** The refuted predicate is the strongest one; the consumer runs on either of the two survivors. A strong-looking lemma was being carried, unmeasured, for a job it was never needed for.
2. **A single instance suffices.** Both branches consume the predicate only at n_0 , never as a universal statement. This matters for anyone trying to prove one: the target is an instance at a hypothetical first failure, not a theorem about all n .
3. **The empirical situation is unchanged and is not a proof.** Over the primes below 10^9 — 50 847 533 indices — (P6'-gov), (P6'-min), (P6'-rec) and $T_{\mu(n)} \leq T_{r(n)}$ all hold with **zero** exceptions, (P6'-rec) across 29 record steps (§9). So the hypothesis of Proposition 4.7 is empirically available on the swept range. That is the honest statement and it is not a proof.

Remark 4.12 (Which margin decays, and which does not). The two survivors behave very differently numerically, and conflating them was the second half of round 1’s collision. The margin of (P6'-gov) *decays* like $p^{-0.83}$: its minimum is $+1.04641539561833 \cdot 10^{-2}$ at $3 \cdot 10^6$ and $+2.93 \cdot 10^{-6}$ at 10^{11} , with zero exceptions throughout — which is why a `float64` check of it approaches its noise floor near the published frontier. The margin of (P6'-min) is *flat*: its global minimum is $+0.48452773339831 \dots$, attained at $n = 1879$ (with $\mu = 1831$) and never approached again across eleven decades. The upstream noise-floor alarm is therefore correct about (P6'-gov) and *does not apply* to (P6'-min), which stands some 10^{12} ulps clear of zero at every decade measured; and the inference sometimes drawn from it — “the route must therefore be the analytic one” — does not survive the rescoping.

5 The monotone-bar principle and the maximal-gap reduction

Section 4 settled what the pruning obligation *is*. This section shows that a verification over a range does not have to wait for it: the pruning can be run against a bar that is monotone by construction, and then the oscillation of T is irrelevant.

Lemma 5.1 (Monotone-bar principle; [P]). *Let B be a real-valued function of the prime p , nondecreasing in p . Call k a B -breach if $g_k \geq B(p_k)$. If a B -breach exists, then the least one, k_0 , satisfies $g_m < g_{k_0}$ for every $m < k_0$; that is, k_0 is a strict maximal-gap index.*

Proof. Idea: Minimality gives an upper bound at every earlier index; monotonicity moves that bound forward to k_0 ; the two chain.

Let $m < k_0$. Then $p_m < p_{k_0}$, so $B(p_m) \leq B(p_{k_0})$ by monotonicity. By minimality of k_0 , the index m is not a B -breach, so $g_m < B(p_m)$. Chaining, $g_m < B(p_m) \leq B(p_{k_0}) \leq g_{k_0}$. $\square$

Only monotonicity of B is used: no arithmetic, no analytic input, no property of the primes beyond $p_m < p_{k_0}$. The conclusion is strict. The lemma says nothing about the *second* breach, and it should not: breaches after the first need not be records. Note also what it does *not* do: with $B = T$ its hypothesis fails outright, since T is not nondecreasing — it decreases at about 55.9% of steps below $3 \cdot 10^6$ (§9). That is exactly why §4 was needed and why this section instantiates B at a surrogate instead.

Explicit bars have validity ranges, and the first few primes sit outside them, so we need a truncated form.

Lemma 5.2 (Truncated monotone-bar principle; [P]). *Let B be nondecreasing and let $1 \leq N_1 \leq N_2$. Suppose*

- (i) $g_m < B(p_m)$ for every maximal-gap index m with $N_1 \leq m \leq N_2$;
- (ii) $\max\{g_m : m < N_1\} < B(p_{N_1})$.

Then $g_k < B(p_k)$ for every k with $N_1 \leq k \leq N_2$.

Proof. Idea: Run Lemma 5.1 inside the window, and let hypothesis (ii) absorb everything below it in one step.

Suppose not, and let k_0 be least in $[N_1, N_2]$ with $g_{k_0} \geq B(p_{k_0})$. Take any $m < k_0$. If $m \geq N_1$, minimality gives $g_m < B(p_m) \leq B(p_{k_0}) \leq g_{k_0}$. If $m < N_1$, then by (ii) and monotonicity $g_m \leq \max\{g_j : j < N_1\} < B(p_{N_1}) \leq B(p_{k_0}) \leq g_{k_0}$. Either way $g_m < g_{k_0}$, so k_0 is a maximal-gap index in $[N_1, N_2]$ breaching B , contradicting (i). $\square$

5.1 Instantiating at Kourbatov's monotone surrogate

Put

$$S(x) := \log^2 x - \log x - 1.17.$$

Lemma 5.3 ([P]). *S is strictly increasing on $x > e^{1/2}$, hence at every prime.*

Proof. Idea: One derivative.

$$dS/dx = (2\log x - 1)/x > 0 \text{ for } \log x > 1/2, \text{ and } e^{1/2} = 1.6487 \dots < 2. \quad \square$$

Lemma 5.4 (S lies below the true bar; [P]). *For every n with $p_n \geq 2\,634\,800\,823$ we have $T_n > S(p_n)$.*

Proof. Idea: Replace the index n by an upper bound on $\pi(p_n)$, then use $e^u - 1 > u$.

By (4), $n = \pi(p_n)$. Axler's corollary on upper bounds for π , in the corrigendum-corrected form, states $\log x - 1 - 1.17/\log x < x/\pi(x)$ for $x \geq 2\,634\,800\,823$ [Axl14]. With $x = p_n$ and $\ell = L_n$ this gives $\pi(x) < x/(\ell - 1 - 1.17/\ell)$, hence $u := \ell/\pi(x) > (\ell^2 - \ell - 1.17)/x$, and therefore $T_n = x(e^u - 1) > xu > \ell^2 - \ell - 1.17 = S(p_n)$. $\square$

Caveat 5.5 (The Axler dependency, as it now stands). Round 1 carried Lemma 5.4 at [P·s] because [Ax14] had not been opened; it was quoted through Kourbatov’s proofs. **It is now opened:** the preprint (arXiv:1409.1780v3), the published version (*Integers* **16** (2016), A22) and the 2018 corrigendum were all fetched, digest-pinned and read at the locator. The mathematical tier improves accordingly and this lemma is stated at [P]. Two purely editorial hazards replace the old one, and neither is decorative.

- (a) **The corollary numbering differs by one between the editions.** The upper-bound corollary is Cor. 3.5 in the preprint and Cor. 3.4 (p. 8) in the journal; the lower-bound table is Cor. 3.6 and Cor. 3.5 respectively. The corrigendum — which moves the validity range from $x \geq 5.43$ to $x \geq 2\,634\,800\,823$, nine orders of magnitude — is written against the *journal* numbering. Both editions point at the same inequality, so no mathematical error propagates, but a locator naming a corollary number without naming its edition matches no single edition. We therefore name the statement rather than a number.
- (b) **One row of the lower-bound table exists in the preprint only.** The row $(a, b, c, d) = (1, 0, 0, 0)$ with $x_0 = 1\,772\,201$ appears in the preprint’s 14-column table and is absent from the journal’s 12-column one. Theorem 6.15 is deliberately built on the $(2.1, 0, 0, 0)$ row with $x_0 = 6\,690\,557$, which is present in *both* editions and is strictly stronger. See Remark 6.17.

The validity range remains load-bearing rather than decorative: the pre-corrigendum range is not merely unproved but false, with 4987066 counterexamples below 10^8 , the smallest at $p = 59\,753$ — all of them below the corrected threshold, which is therefore corroborated rather than contradicted. Correspondingly, in-run data below $3 \cdot 10^6$ exhibits 14426 indices with $T_n \leq S(p_n)$, so Lemma 5.4 is false outside its stated range.

Theorem 5.6 (Maximal-gap reduction; [P]). *Let $N_2 \geq 10$ and suppose $g_m < S(p_m)$ for every maximal-gap index $m \in [10, N_2]$. Then $g_k < S(p_k)$ for every $k \in [10, N_2]$, and consequently F holds at every $k \in [10, N_2]$ with $p_k \geq 2\,634\,800\,823$.*

Proof. Idea: Lemma 5.2 with the monotone surrogate S in place of the oscillating T ; the small-index hypothesis is one finite comparison.

Apply Lemma 5.2 with $B = S$ (nondecreasing by Lemma 5.3), $N_1 = 10$ and N_2 as given. Hypothesis (i) is the assumption. Hypothesis (ii) is the finite fact $\max\{g_j : j \leq 9\} = 6 < 6.80139\dots = S(29) = S(p_{10})$, verified in exact arithmetic (§9, item V2). Hence $g_k < S(p_k)$ throughout $[10, N_2]$. Where $p_k \geq 2\,634\,800\,823$, Lemma 5.4 gives $S(p_k) < T_k$, so $g_k < T_k$, which is F at k by Lemma 2.1. $\square$

Remark 5.7 (What this settles about the obligation). Theorem 5.6 discharges the entire practical content of the pruning obligation without proving any of the four predicates of Definition 4.1. The oscillation of T is irrelevant to the pruning *as it is actually run*, because it is run against S , which does not oscillate. The pieces were all present in the attack’s own concept-card set — the bar S , the (unstated but immediate) monotonicity of S , and the open obligation — and were never joined. The join is Lemma 5.2. What the predicates are still needed for is the *exact* bar: Theorem 4.11 is a statement about T , and no surrogate substitutes for it there.

Remark 5.8 (Range discipline). Theorem 5.6 delivers nothing below $p_k = 2\,634\,800\,823$, where Lemma 5.4 is unavailable and in fact false. That segment is covered by direct computation, i.e. by Theorem 2.3, and the two must be joined explicitly, never assumed to overlap.

Remark 5.9 (The Dusart-only fallback). The Axler dependency of Lemma 5.4 can be removed entirely at the cost of a looser bar: Lemma 6.2 below proves $T_n > B(p_n)$ with $B(x) =$

$\log^2 x - 1.1 \log x$ for $p_n \geq 60184$, from Dusart alone, and B is monotone by Lemma 6.3. Substituting B for S throughout Theorem 5.6 gives a version resting on a single L0 source, with a bar looser by roughly $0.1L$ — a relative deficit measured at 0.193% at $p = 2 \cdot 10^7$ and 0.196% at $p = 1.69 \cdot 10^{15}$. We state the Axler version because it is the one the literature uses; we flag the fallback because it is the one that needs the fewest sources.

6 The unconditional finite-range architecture

The published frontier 2^{64} is not a per-pair sweep: it is a theorem about a table. This section reconstructs that theorem from first principles, with explicit constants, resting on *one* cited inequality — and in particular avoiding the Axler chain of Caveat 5.5 entirely.

6.1 The analytic input

Lemma 6.1 (Dusart’s upper bound for π ; cited, tier L0). $\pi(x) \leq x/(\log x - 1.1)$ for all $x \geq 60184$.

This is [Dus10, Thm. 6.9, eq. (6.6)], read at the locator. It is the only external mathematical input to Theorem 6.4.

Lemma 6.2 (Explicit floor under the bar; [P]). Set $B(x) := \log^2 x - 1.1 \log x$. Then $T_n > B(p_n)$ for every n with $p_n \geq 60184$.

Proof. Idea: The only opaque ingredient of T_n is the index $n = \pi(p_n)$; an upper bound on π turns the bar into a lower bound in $\log p_n$ alone.

Since $e^t - 1 > t$ for $t > 0$ and $L_n/n > 0$, we have $T_n > p_n L_n/n$. By (4), $n = \pi(p_n)$, and Lemma 6.1 at $x = p_n \geq 60184$ gives $n \leq p_n/(L_n - 1.1)$, with $L_n - 1.1 > 0$ in this range. Hence $p_n/n \geq L_n - 1.1$ and

$$T_n > L_n \cdot \frac{p_n}{n} \geq L_n(L_n - 1.1) = B(p_n). \quad \square$$

The direction is the content. T_n decreases in n at fixed p_n , so lower-bounding T_n requires an upper bound on the rank, which is precisely what Dusart’s upper bound on π supplies. The validity range is load-bearing and not decoration: $L(L - 1.1)$ crosses $L^2 - L - 1$ — the true asymptotic size of T_n — exactly at $L = 10$, i.e. $x = e^{10} = 22026$, below which Lemma 6.2 is false. Dusart’s threshold 60184 clears that crossing, but only by about 0.10 in L .

Lemma 6.3 ([P]). B is strictly increasing on $[e^{0.55}, \infty) \supseteq [2, \infty)$.

Proof. Idea: One derivative, in the variable L .

With $L = \log x$, $dB/dL = 2L - 1.1 > 0$ for $L > 0.55$, and $L \mapsto \log x$ is strictly increasing; $e^{0.55} = 1.7333 < 2$. $\square$

Lemma 6.3 is the whole point: B is a function of p alone, not of the pair (p_n, n) , so the non-monotone object T never appears in a comparison between two different indices. In particular this section is independent of all four predicates of Definition 4.1, and any downstream argument that gates the verified range on one of them is importing a dependency that is not there.

6.2 The finite-range theorem

Theorem 6.4 (Unconditional finite-range verification; [P]). *Let $X_0 := 60\,184$ and let $X \geq X_0$. Suppose:*

- (H1) $p_{n+1}^n < p_n^{n+1}$ for every n with $p_n < X_0$;
- (H2) a table is available listing, for every gap value g occurring as $g_n = g$ for some n with $p_n \leq X$, the first occurrence $q(g) := \min\{p_n : g_n = g\}$;
- (H3) for every gap value g in that table with $q(g) \geq X_0$, one has $g \leq B(q(g))$.

Then **F** holds at every n with $p_n \leq X$.

Proof. Idea: A gap that was already safe where it first occurred is safe everywhere later, because the floor B only rises; gaps whose first occurrence is below X_0 are all too small to matter.

Let n satisfy $p_n \leq X$. If $p_n < X_0$, then (H1) and Lemma 2.1 give **F** at n . Otherwise $p_n \geq X_0$. Put $g := g_n$ and $q := q(g) \leq p_n$.

If $q \geq X_0$: by (H3), $g \leq B(q)$; by Lemma 6.3 and $q \leq p_n$, $B(q) \leq B(p_n)$; by Lemma 6.2, $B(p_n) < T_n$. Hence $g_n < T_n$, and Lemma 2.1 gives **F** at n .

If $q < X_0 \leq p_n$: the first occurrence of g lies in the base-case range, so $g \leq G_0 := \max\{g_k : p_k < X_0\} = 72$, attained at $p = 31\,397$ (§9, item V1, exact integer arithmetic over the 6076 gaps below X_0). Since $B(X_0) = B(60\,184) = 109.008\dots > 72 \geq g$, Lemmas 6.3 and 6.2 give $g \leq G_0 < B(X_0) \leq B(p_n) < T_n$, so **F** at n . $\square$

Corollary 6.5 (Decision procedure; [P]). *For a gap value g define*

$$S^\sharp(g) := \exp\left(\frac{1.1 + \sqrt{1.21 + 4g}}{2}\right),$$

*the unique p with $B(p) = g$, so that $g \leq B(p) \iff p \geq S^\sharp(g)$. Then hypothesis (H3) reads $q(g) \geq S^\sharp(g)$, and Theorem 6.4 says: a gap of size g occurring at a prime $p \geq \max(S^\sharp(g), 60\,184)$ cannot violate **F**.*

Proof. Idea: Invert the quadratic $B(p) = \log^2 p - 1.1 \log p$ in the variable $\log p$ and use Lemma 6.3.

Solving $L^2 - 1.1L = g$ for $L > 0$ gives $L = (1.1 + \sqrt{1.21 + 4g})/2$; B is strictly increasing (Lemma 6.3), so $g \leq B(p)$ iff $\log p$ exceeds that root. The restatement of (H3) is immediate, and the displayed sentence is the second case analysis of Theorem 6.4 read backwards. $\square$

Note that a gap size which passes is safe *everywhere*, not merely below X : $S^\sharp(g)$ does not depend on X . The verification of a range therefore costs one finite integer check below 60 184, plus one comparison per distinct gap value. Nothing else.

6.3 Independent reproduction of the published constant 1920

Proposition 6.6 ([C]). $B(2^{64}) = L(L - 1.1)|_{L=\log 2^{64}} = 1919.13798349753288\dots$ (with $\log 2^{64} = 44.36141955583649\dots$), so a gap of at least 1920 is required to violate **F** at a prime just below 2^{64} . Equivalently, the largest even g with $S^\sharp(g) \leq 2^{64}$ is $g = 1918$: $S^\sharp(1918) = 1.8209 \cdot 10^{19} \leq 2^{64} = 1.8447 \cdot 10^{19} < S^\sharp(1920) = 1.8629 \cdot 10^{19}$.

Kourbatov’s 2023 endnote states that “prime gaps of size $g < 1920$ cannot violate” the conjecture [Kou15b]; since every prime gap after $g_1 = 1$ is even, $g < 1920$ means exactly $g \leq 1918$. The published integer falls out of Lemma 6.2 with no tuning.

Caveat 6.7 (What the agreement is worth, and what it is not). Three caveats, all stated rather than glossed. (a) The two derivations are not independent in their analytic ingredient: both replace n by an upper bound on $\pi(p_n)$; they differ in *which* bound (Dusart eq. (6.6) here, the Axler-based $L^2 - L - 1.17$ there). What the agreement does establish is that no sign error, inverted inequality or transcription slip separates the two chains. (b) Under Kourbatov's sharper constant the same computation yields 1922, so his published $g < 1920$ is conservative relative to his own criterion, and the exact coincidence with 1918 is partly arithmetic luck at even-gap granularity. (c) Lemma 6.2 yields the *local* statement at the frontier; the endnote's phrasing is *global*. Since $S^\sharp(1918) = 1.82 \cdot 10^{19}$, our bound leaves a gap of 1918 free to violate F anywhere below $1.82 \cdot 10^{19}$. Closing that needs the first-occurrence table again.

6.4 The table-free window, and its permanent closure

Can a range be certified with *no* prime enumeration at all? Only just, and only once.

Proposition 6.8 (Table-free window; [P]). *F holds at every n with $396\,738 \leq p_n \leq 777\,600$, with no computational input beyond two cited explicit estimates, both from [Dus10].*

Proof. Idea: Dusart's short-interval theorem caps g_n by $p/(25L^2)$; Lemma 6.2 floors T_n by $L^2 - 1.1L$; the two cross exactly once.

For $p_n \geq 396\,738$, [Dus10, Prop. 6.8] supplies a prime in $(x, x(1 + 1/(25 \ln^2 x)))$, hence $g_n \leq p_n/(25L_n^2)$. By Lemma 6.2, $T_n > L^2 - 1.1L$. So it suffices that

$$\frac{p}{25L^2} \leq L^2 - 1.1L \quad \Longleftrightarrow \quad p \leq 25L^3(L - 1.1), \quad L = \log p.$$

Let $h(p) := p - 25(\log p)^3(\log p - 1.1)$. Then $h(396\,738) = -2.35 \cdot 10^5 < 0$, and $h'(p) = 1 - (25/p)(4L^3 - 3.3L^2) > 0$ throughout $[4 \cdot 10^5, \infty)$, so h has a unique sign change there, at $p^* = 777\,600.744 \dots$ (§9, item V6). Hence the displayed inequality holds on $[396\,738, p^*]$, so $g_n < T_n$, so F at n by Lemma 2.1. $\square$

Proposition 6.9 (The window closes permanently; [P]). *For $p > p^* = 777\,600.744 \dots$, [Dus10, Prop. 6.8] does not imply F, and the shortfall grows without bound: $\frac{p/(25L^2)}{L^2 - 1.1L} = \frac{p}{25L^3(L - 1.1)} \rightarrow \infty$.*

Proof. Idea: p outgrows every fixed power of $\log p$.

Immediate from the display in the proof of Proposition 6.8. $\square$

Remark 6.10 (The precise measure of what analysis alone can do). A table-free proof of F on $[X_0, X]$ requires an unconditional explicit gap bound of quality $g_n = O(\log^2 p_n)$ on that range. Every known unconditional gap bound — Dusart's $p/(25 \log^2 p)$ here, $p^{0.525}$ of [BHP01], and everything between — is a *power of p* . So the table-free route is dead above $7.776 \cdot 10^5$ and will stay dead until unconditional prime-gap technology reaches $\log^2$ scale, which is exactly the strength that proving F itself requires. **Corollary, stated so that it is not re-derived: no amount of analysis will ever eliminate the computational input from a verified-range claim.** The verified range is irreducibly a computation plus a theorem about the computation. Theorem 6.4 is the theorem; §11.4 names the computation. The window $[396\,738, 777\,600]$ lies entirely above X_0 and is therefore already covered by (H1)–(H3) in any real verification; its value is that it is the exact measure of how much analysis alone can do, and the measure is: one factor of about 2 in p , once, at $p \approx 10^6$.

6.5 Near-record maximality against the exact bar

Lemma 5.1 does not settle (M1) against the *exact* bar T , because T is not monotone, and Theorem 4.11 settles it only from a predicate nobody has proved. What can be proved outright is a windowed version: the first failure's gap exceeds every gap occurring below a definite multiple of p_{n_0} . Round 1 stated one such theorem and withheld a second because its derivation was broken. Round 2 repaired the broken one — twice, in two independent ways — and tightened the first. Both repaired forms are stated here; the sharper one is carried and the other is retired to Remark 6.17.

The engine is the following sandwich, which uses no monotonicity at all.

Lemma 6.11 (Sandwich; [P]). *Suppose $A(x) \leq T_n \leq C(x)$ at $x = p_n$ for every n with $p_n \geq X_0$, and let n_0 be the least failure of F . Then for every $m < n_0$ with $p_m \geq X_0$ and $C(p_m) \leq A(p_{n_0})$, one has $g_m < g_{n_0}$.*

Proof. Idea: Transport the bar from m to n_0 through the two envelopes, then use minimality at m and failure at n_0 .

$T_m \leq C(p_m) \leq A(p_{n_0}) \leq T_{n_0}$. Since $m < n_0$ and n_0 is least, m is not a failure, so $g_m < T_m \leq T_{n_0} \leq g_{n_0}$. $\square$

Two elementary facts convert π -estimates into envelopes for T .

Lemma 6.12 ([P]). *For $x > 0$ and $t > 0$: (E1) $x(e^{t/x} - 1) > t$; and (E2) if $0 < t \leq x$ then $x(e^{t/x} - 1) \leq t(1 + t/x)$.*

Proof. Idea: (E1) is $e^s > 1 + s$; (E2) sums the tail of the exponential series and bounds it by $e - 2 < 1$.

(E1) is immediate from $e^s > 1 + s$ at $s = t/x$. For (E2), write $x(e^{t/x} - 1) = t + t \sum_{k \geq 2} (t/x)^{k-1}/k!$; for $0 < t/x \leq 1$ the sum is at most $\sum_{k \geq 2} 1/k! = e - 2 = 0.71828 \dots < 1$. $\square$

Write $x = p_n$, $\ell = L_n$ and $u = \ell/\pi(x)$, so that $T_n = x(e^u - 1)$ exactly by (4). The four envelopes used below are then:

	Envelope	Range	From
(D-low)	$T_n > \ell^2 - 1.1\ell$	$x \geq 60\,184$	[Dus10, Thm. 6.9, (6.6)], upper half; (E1)
(D-high)	$T_n \leq v(1 + v/x)$, $v := \ell^2 - \ell$	$x \geq 5393$	[Dus10, Thm. 6.9, (6.6)], lower half; (E2)
(A-low)	$T_n > \ell^2 - \ell - 1.17$	$x \geq 2\,634\,800\,823$	Axler, upper-bound corollary, as corrected
(A-high')	$T_n \leq v(1 + v/x)$, $v := \ell^2 - \ell - 1 - \frac{2.1}{\ell}$	$x \geq 6\,690\,557$	Axler, table row (2.1, 0, 0, 0); (E2)

Caveat 6.13 (The round-1 defect, stated precisely because a constant depended on it). Round 1's corresponding upper envelope was printed as $T_n \leq (\ell^2 - \ell - 1 - 1/\ell)(1 + \ell^4/x)$, justified as " $T_n \leq v(1 + v/x) \leq v(1 + \ell^4/x)$ using $v < \ell^2$ ". The step $v(1 + v/x) \leq v(1 + \ell^4/x)$ is true but weakens the bound by an additive $v(\ell^4 - v)/x \approx \ell^6/x$, not the v^2/x that the algebra consuming it silently assumed. **As printed the bound does not follow from its stated justification, and the requirement it induces is false by a factor 38.8137468954... at the bottom of its range:** the true required separation there is 0.169339812744..., not the 0.004479 printed. The in-run numerical check reproduced the error rather than catching it, which is the general lesson — *a check written from a derivation cannot catch an error in the derivation*. The repair is not to re-insert ℓ^4 correctly; it is never to introduce ℓ^4 at

all, which is what (D-high) and (A-high') do. Round 1's *conclusion* survives: the printed constant 0.004479 is in fact sufficient, but for a reason the printed derivation did not supply, and both round-2 repairs below establish it independently.

Throughout the rest of this subsection, n_0 is the least failure of F. By Theorem 2.3, $p_{n_0} > 2^{64}$, so $\lambda := L_{n_0} > 44.36141955\dots$ and, by (D-low), $T_{n_0} > \lambda^2 - 1.1\lambda > 1919.1379834\dots$, whence $g_{n_0} \geq T_{n_0} > 1919$.

Theorem 6.14 (Near-record maximality, Dusart only; [P·s]). *Let $m < n_0$. If $p_m \leq p_{n_0}e^{-0.0516}$, i.e. $p_m \leq 0.94970p_{n_0}$, then $g_m < g_{n_0}$.*

Proof. Idea: Below 10^8 every gap is too small to matter; above it, ask how far apart the two logarithms must be for (D-high) at m to fall below (D-low) at n_0 , and bound the answer by a majorant on a partition rather than by sampling.

Small branch. If $p_m < 10^8$ then $g_m \leq 220$ (the largest prime gap below 10^8 , attained at $p = 47\,326\,693$; §9, item V10) and $220 < 1919 < g_{n_0}$.

Main branch. Let $p_m \geq 10^8$, so $\ell := L_m \geq \ell_D := \log 10^8 = 18.4206807\dots$, and both (D-high) at m and (D-low) at n_0 apply. Put $d := \lambda - \ell \geq 0$, $v := \ell^2 - \ell$ and $\varepsilon := v^2/p_m$. By Lemma 6.11 it suffices that $v(1 + v/p_m) \leq \lambda^2 - 1.1\lambda$. Expanding $\lambda = \ell + d$,

$$\lambda^2 - 1.1\lambda - v = 2\ell d + d^2 - 1.1d - 0.1\ell,$$

so it suffices that $d(2\ell - 1.1) \geq 0.1\ell + \varepsilon$, i.e. that $d \geq d^*(\ell) := (0.1\ell + \varepsilon)/(2\ell - 1.1)$.

Since $p_m \geq e^\ell$ we have $\varepsilon(\ell) \leq (\ell^2 - \ell)^2 e^{-\ell} \leq \ell^4 e^{-\ell}$, and $\ell \mapsto \ell^4 e^{-\ell}$ is decreasing for $\ell > 4$. Partition $[\ell_D, \infty)$. On a cell $[a, b]$ we have $0.1\ell \leq 0.1b$, $\ell^4 e^{-\ell} \leq a^4 e^{-a}$ and $1/(2\ell - 1.1) \leq 1/(2a - 1.1)$, so

$$\sup_{[a,b]} d^*(\ell) \leq \frac{0.1b + a^4 e^{-a}}{2a - 1.1}.$$

This is a *majorant*, not a *sample*, so a maximum over a partition is a proof. Over $[\ell_D, 1000]$ in cells of width 0.01 the majorant is largest on the first cell, where it equals 0.051599027. For $\ell \geq 1000$, $0.1\ell/(2\ell - 1.1)$ decreases to 0.05, so the majorant there is at most $(100 + 1000^4 e^{-1000})/1998.9 = 0.050028$. Hence $d^* \leq 0.051599 < 0.0516$ throughout, and $e^{-0.0516} = 0.949708674346\dots$ $\square$

Theorem 6.15 (Near-record maximality with Axler; [P·s]). *Let $m < n_0$. If $p_m \leq p_{n_0}e^{-0.0017569}$, i.e. $p_m \leq 0.998244p_{n_0}$, then $g_m < g_{n_0}$.*

Proof. Idea: The same majorant argument, with (A-high') at m and (A-low) at n_0 ; the surplus to be beaten is now $0.17 - 2.1/\ell$ rather than 0.1ℓ , which is bounded rather than growing — that is the whole gain.

Small branch. If $p_m < 6\,690\,557$ then $g_m \leq 154$ (the largest prime gap below $6\,690\,557$, attained at $p = 4\,652\,353$; §9, item V10) and $154 < 1919 < g_{n_0}$.

Main branch. Let $p_m \geq 6\,690\,557$, so $\ell := L_m \geq \ell_A := \log 6\,690\,557 = 15.7162077\dots$ and (A-high') applies at m ; (A-low) applies at n_0 because $p_{n_0} > 2^{64} > 2\,634\,800\,823$. With $v := \ell^2 - \ell - 1 - 2.1/\ell$ and $\lambda = \ell + d$, Lemma 6.11's hypothesis $v(1 + v/p_m) \leq \lambda^2 - \lambda - 1.17$ expands to

$$d(2\ell - 1) + d^2 \geq 0.17 - \frac{2.1}{\ell} + \frac{v^2}{p_m} =: \psi(\ell),$$

so $d \geq \psi(\ell)/(2\ell - 1)$ suffices, and $d = 0$ suffices wherever $\psi(\ell) \leq 0$. As before $v^2/p_m \leq \ell^4 e^{-\ell}$, decreasing; $0.17 - 2.1/\ell$ is increasing; $1/(2\ell - 1)$ is decreasing. So on a cell $[a, b]$,

$$\sup_{[a,b]} \frac{\psi(\ell)}{2\ell - 1} \leq \frac{0.17 - 2.1/b + a^4 e^{-a}}{2a - 1}.$$

Over $[\ell_A, 300]$ in cells of width 0.01 this majorant is largest on the cell beginning at $\ell = 24.40621$, where it equals 0.0017568759; for $\ell \geq 300$ it is at most $(0.17 + 300^4 e^{-300})/599 = 0.00028381$. Hence $d \geq 0.0017569$ suffices, and $e^{-0.0017569} = 0.998244642445 \dots$ $\square$

Remark 6.16 (Round 1 against round 2, and the sliver that remains).

branch	round 1	round 2
Dusart only	$d \geq 0.0623, p_m \leq 0.93961 p_{n_0}$	$\mathbf{d} \geq \mathbf{0.0516}, \mathbf{p_m} \leq \mathbf{0.94970 p_{n_0}}$
with Axler	$d \geq 0.004479, p_m \leq 0.99553 p_{n_0}$ (from a lemma that did not support it)	$\mathbf{d} \geq \mathbf{0.0017569}, \mathbf{p_m} \leq \mathbf{0.998244 p_{n_0}}$

The Dusart branch rests on an L0 source in both rounds; the Axler branch rested on an unopened source in round 1 and on one opened in both editions in round 2. The Dusart branch improves only because the small-branch cutoff rises from 60 184 to 10^8 , which is licensed by $g_m \leq 220 < 1919$; it **cannot** improve much further, since $d^*(\ell) \rightarrow 0.05$, so the Dusart-only sliver is pinned near 5% at every scale. The residual sliver on the Axler branch has relative width 0.176% at 2^{64} . Two sanity anchors, reported so the repair is not overstated: solving the quadratic exactly rather than majorising gives 0.051493457 and 0.0017560603, both *below* the constants proved above, as a majorant requires.

Remark 6.17 (The other repair, and why it is retired). A second round-2 leg repaired the same round-1 defect independently and correctly, into a different theorem: with $v := \ell^2 - \ell - 1 - 1/\ell$ in place of (A-high')'s v — the tight restatement of round 1's own printed lemma — and a proof that the majorant is decreasing (so that its supremum is attained at the left endpoint, 0.004363568), one obtains $d \geq 0.0043636$, i.e. $p_m \leq e^{-0.0043636} p_{n_0} = 0.995645906670 \dots p_{n_0}$. That statement is verified correct, and it is the cleanest available proof that round 1's *conclusion* survived its own broken derivation, since 0.004363568 is below the printed 0.004479. **We nevertheless do not carry it**, for a reason that is not a matter of taste: it consumes the Axler table row (1, 0, 0, 0) with $x_0 = 1\,772\,201$, which exists in the preprint edition only (Caveat 5.5(b)), whereas Theorem 6.15's row is present in both. Two correct repairs of one defect, shipped by two legs that did not cite each other, is itself one of the corpus's unrec-onciled seams; §11.3 records it as such rather than presenting the choice as if the corpus had made it.

Caveat 6.18 (What “unconditional” means for Theorem 6.14). Theorem 6.14 uses no analytic source beyond [Dus10], and it is sometimes labelled “unconditional” on that ground. The label is too strong as usually written, and the honest form is: *unconditional given the published 2^{64} verification height and a finite in-run gap computation*. Both are named inputs; neither is an analytic hypothesis; but the first is Theorem 2.3, which rests on a source this work never opened, and the second is our own sieve. **That is why both Theorems 6.14 and 6.15 carry [P·s] here and not [P]**, even though their own analytic inputs are at the ledger's top tier: the weakest link is the verification height, not the estimate. Round 1 was faulted for exactly this pattern in a different branch, and it reappeared in round 2 in the branch round 1 had passed clean; §11.3 records it.

6.6 The obstruction inside the residual window

Theorems 6.14 and 6.15 leave a sliver: the primes p_m within a factor e^{-d} of p_{n_0} . The obstruction there is exact and worth naming, because it explains why no sharpening of constants closes the window.

To first order in y/p_m , with $y = p_n - p_m$ and $k = \pi(p_n) - \pi(p_m)$,

$$T_m \leq T_n \iff k \leq y \left(1 + \frac{1}{L_m}\right) \frac{\pi(p_m)}{p_m} (1 + O(y/p_m)),$$

i.e., since $\pi(x)/x \approx 1/(\log x - 1)$: $T_m \leq T_n$ holds exactly when the interval $(p_m, p_n]$ contains no more than about $y/(L-2)$ primes. The surplus $1/(L-1) - 1/L \approx 1/L^2$ between the running-average density and the local density is the entire drift that makes T coarsely increase.

Proposition 6.19 (Obstruction; [P]). *Closing the residual window requires an unconditional upper bound on the number of primes in an interval of length y of order $x/\log x$ that is within a factor $1 + 2/L$ of the truth. The unconditionally available bound of Brun–Titchmarsh type gives only a factor of about 2.*

The required relative slack over the prime-number-theorem count is 4.7% at 2^{64} and shrinks to 1% at $L = 200$, while the Brun–Titchmarsh-to-needed ratio stays near 2.1 across that range. So the needed estimate is not a constant improvement but an asymptotically exact short-interval count. Independently, a computational route to the same wall prices it: typical windows are settled unconditionally by Brun–Titchmarsh at 99.861% of governed indices, but the extremal configuration needs a short-interval count sharp to $1 + 2.2/\log p$ — Cramér strength — and that gap *widens* with scale even as empirical coverage improves. A pruning rule is worth its worst case. *This is, in our judgement, the most tractable genuinely open analytic node the attack isolated — and it is still an open problem in analytic number theory, not a lookup.*

Caveat 6.20 (Unsourced, and used only negatively). Neither the Brun–Titchmarsh inequality nor the second Hardy–Littlewood conjecture is covered by a source-ledger row in this work. Both are standard and undisputed; they are used here only to *name* an obstruction, never to support a positive claim. Proposition 6.19 is stated at [P] for its criterion $(1 + 2/L$, derived above) and its comparison to Brun–Titchmarsh is flagged as unsourced.

7 The Riemann-Hypothesis route is closed

The slogan “RH does not help with Firoozbakht” is common and usually unargued. This section replaces it with theorems. Five inequivalent readings of “an RH-conditional bound settles F” are separated; four are refuted, and the fifth is left undecided with a lower bound on the strength of what a proof would have to yield.

Throughout this section we use the CMS envelope

$$B_n := \frac{22}{25} \sqrt{p_n} L_n,$$

for which [CMS19, §1.2, Thm. 5] gives: RH implies $g_n \leq B_n$ for every $n \geq 3$. The hypothesis is stated there as $p_n > 3$, and [Vis18, Thm. 1, eq. (1.4)] states the same bound with hypothesis $n \geq 3$, $p_n \geq 5$; over the primes $p_n > 3 \iff p_n \geq 5 \iff n \geq 3$, so these are the same hypothesis. **That hypothesis is part of the statement and is carried in every theorem below.** (The two sources differ in strictness, $\leq$ against $<$; every argument here consumes the bound only as a non-strict upper envelope, so the discrepancy is inert here, and it is recorded rather than smoothed.)

We say the envelope *certifies* F at n when $B_n \leq T_n$; by Remark 2.2 this upgrades to the strict inequality $g_n < T_n$ that Lemma 2.1 demands.

We also use the following in-run fact, verified for all $n \leq 216\,815$ and asymptotically a consequence of [Dus10, Thm. 6.9]: $\{n : T_n \geq L_n^2\} = \{1, \dots, 7, 10\}$.

Lemma 7.1 ([P]). *For every real $x > 0$, $\sqrt{x} > \frac{25}{22} \log x$; equivalently $\frac{22}{25} \sqrt{x} \log x > (\log x)^2$ for all $x > 1$. Consequently $B_n > L_n^2$ at every index $n \geq 1$.*

Proof. Idea: One stationary point, and it sits above zero.

Put $k := 25/22$ and $h(x) := \sqrt{x} - k \log x$ on $(0, \infty)$. Then $h'(x) = (\sqrt{x} - 2k)/(2x)$, so h has a unique stationary point at $x^* = (2k)^2 = 625/121 = 5.1652892561983471\dots$, a minimum, with $h \rightarrow +\infty$ at both ends. At that point

$$h(x^*) = \frac{25}{11} - \frac{25}{22} \log \frac{625}{121} = 0.40686238165947680\dots > 0.$$

Hence $h > 0$ throughout. Multiplying $\sqrt{x} > k \log x$ by $\log x > 0$ (valid for $x > 1$) and dividing by k gives the second form; at $x = p_n \geq 2 > 1$ it gives $B_n > L_n^2$. The last claim holds at every index because B_n is just a function of p_n : the CMS hypothesis plays no role in it. That is precisely why the arithmetic clearance set of Theorem 7.2 can be, and is, larger than the certification set of Theorem 7.3. $\square$

7.1 The certification set, with its quantifier attached

Round 1's headline for this section read "the CMS envelope certifies F at exactly one index and at no other index whatsoever". **The second clause is false as stated**, and round 2 proves the two statements separately rather than patching the prose. The arithmetic and the hypothesis do different work, and the theorem that mixes them cannot say which did what.

Theorem 7.2 (Arithmetic clearance; no hypothesis, no RH; [P]). $A := \{n \geq 1 : B_n < T_n\} = \{1, 2, 3\}$, and the same set is obtained with $\leq$ in place of $<$.

Proof. Idea: Off a set of eight indices the bar T_n sits below L_n^2 , while the envelope is above L_n^2 always; the eight are checked by hand.

Let $n \geq 1$. If $n \notin \{1, \dots, 7, 10\}$ then $T_n < L_n^2$, while Lemma 7.1 gives $B_n > L_n^2 > T_n$, so $n \notin A$. The eight remaining indices are checked directly in 50-digit arithmetic:

n	p_n	T_n	B_n	$B_n < T_n?$
1	2	2.0000000000	0.86262716625	yes
2	3	2.1961524227	1.6745100256	yes
3	5	3.5498797334	3.1669550684	yes
4	7	4.3860359319	4.5305870087	no
5	11	6.7693369282	6.9985686377	no
6	13	6.9342810812	8.1382896560	no
7	17	8.4816378252	10.279841332	no
10	29	11.610449607	15.957429839	no

All eight comparisons are strict, so $<$ and $\leq$ select the same indices. Hence $A = \{1, 2, 3\}$. $\square$

Theorem 7.3 (The CMS envelope certifies at one index; [P]). Let $S := \{n \geq 3 : B_n \leq T_n\}$ be the set of indices at which the CMS envelope certifies F under RH. Then $S = A \cap [3, \infty) = \{3\}$. That is: the sharpest published RH-conditional prime-gap bound certifies the Firoozbakht inequality at **exactly one index of the range where that bound is available**, namely $n = 3$ ($p_3 = 5$).

Proof. Idea: Intersect Theorem 7.2's clearance set with the source's own hypothesis, and check that the certification at the surviving index is genuine.

$S = A \cap [3, \infty)$ by definition, using the strictness established in Theorem 7.2 so that the $\leq$ in S and the $<$ in A select the same indices. Theorem 7.2 gives $A = \{1, 2, 3\}$, so $S = \{3\}$. At $n = 3$ the certification is genuine: RH gives $g_3 \leq B_3 < T_3$, hence $g_3 < T_3$, hence F at 3 by Lemma 2.1. $\square$

Remark 7.4 (Read the quantifier). Comparing the two theorems: the envelope clears the bar at $n = 1$ and $n = 2$ as well, and what excludes those two indices is **the source’s hypothesis** $p_n > 3$, **not the arithmetic**. So the correct informal form is “at exactly one index of the range where the CMS bound is available”, never “at no other index whatsoever”. This is not a nicety. A finite check that starts where its own theorem starts cannot detect that the range is doing work, and round 1’s check started at $n = 3$ — exactly where its theorem started — which is why the two rows at $n = 1, 2$ were never printed.

Remark 7.5. The single certified index is worthless: $n = 3$ means F at $p = 5$, i.e. $7^3 = 343 < 5^4 = 625$, decidable by hand, and $n = 3$ falls below the threshold $k > 9$ that both Kourbatov criteria carry by hypothesis. Moreover RH is not even load-bearing where the envelope is smallest: by [Vis18, §7] the CMS inequality holds *unconditionally* for all $p < 1.836 \cdot 10^{19}$, essentially the whole verified range of Theorem 2.3 — and what is already known still certifies F at no index above $n = 3$.

7.2 The per-index critical constant, and a seam it closes

Restoring the quantifier did more than correct a sentence: it dissolved a contradiction between two artifacts of this work that had gone unnoticed because both were numerically right and only the words disagreed. One reported that the CMS envelope certifies at “one index and no other”; the other reported a certified range “ $p \leq 5$ ”, i.e. three certified primes $\{2, 3, 5\}$. Both are level sets of a single object.

Definition 7.6 (Per-index critical constant). For $n \geq 1$ put

$$C_n := \frac{T_n}{\sqrt{p_n} L_n},$$

the largest constant C for which the envelope $C\sqrt{p_n}L_n$ still lies at or below the bar at index n .

Then $B_n < T_n \iff C_n > 22/25$, so Theorem 7.2 says exactly that the level set $\{n : C_n > 22/25\}$ is $\{1, 2, 3\}$ — three indices, i.e. the primes 2, 3, 5, i.e. “ $p \leq 5$ ”. The two artifacts were measuring the same level set from opposite sides, one counting indices under the source’s hypothesis and one counting primes without it. They now cite each other through C_n . *This is the one seam in either round that the legs closed by themselves rather than deferring*; §11.3 lists the ones they did not.

7.3 No power-type envelope suffices

Theorem 7.7 ([P]). *Let $C > 0$, $\theta > 0$, $A \in \mathbb{R}$ and $E(x) := Cx^\theta(\log x)^A$. Then $\#\{n : E(p_n) \leq T_n\} < \infty$; indeed $E(p_n)/T_n \rightarrow \infty$.*

Proof. Idea: In the variable $u = \log x$ the envelope is exponential and the bar is quadratic.

Since $T_n < L_n^2$ for all $n \notin \{1, \dots, 7, 10\}$, it suffices that $E(x)/(\log x)^2 \rightarrow \infty$. Write $u := \log x$; then $E(x)/(\log x)^2 = Ce^{\theta u}u^{A-2} \rightarrow \infty$ as $u \rightarrow \infty$, because $e^{\theta u}$ with $\theta > 0$ dominates any fixed power of u . Hence $E(p_n)/T_n > E(p_n)/L_n^2 \rightarrow \infty$, and a divergent ratio is ≤ 1 only finitely often. $\square$

Every prime-gap upper bound currently available lives in this class: $p_n^{0.525}$ unconditionally [BHP01]; $O(\sqrt{p_n} \log p_n)$ under RH (Cramér 1919, as reported in [Vis18, §2, Thm. 4]); $4\sqrt{p_n} \log p_n$ under RH (Goldston 1982, [Vis18, §2, Thm. 5]); the explicit CMS bound and its asymptotic refinement $\limsup \leq 1/C^+(B) < 21/25$ [CMS19, §1.2, Cor. 4]. **The column**

that matters is θ : every entry has $\theta > 0$, while the bar (3) has $\theta = 0$. Improving the exponent from 0.525 to 0.5 — exactly what RH buys — moves nothing across the line, because the line is at $\theta = 0$. Numerically the same statement reads: the slack that Baker–Harman–Pintz supplies overtakes what is needed from $n = 245$ on, and RH’s from $n = 3$ on; both are insufficient exactly where it matters.

Theorem 7.8 (The critical constant is $2/e$; [P]). *For $C > 0$ set $E_C(x) := C\sqrt{x}\log x$. Then $\{x > 1 : E_C(x) \leq (\log x)^2\} \neq \emptyset$ if and only if $C \leq 2/e = 0.7357588823428846\dots$, and when non-empty this set is the bounded interval with endpoints $\exp(-2W_0(-C/2))$ and $\exp(-2W_{-1}(-C/2))$, where W_0, W_{-1} are the two real branches of the Lambert W function. Consequently, for every $C > 0$, E_C certifies F at only finitely many indices.*

Proof. Idea: The condition is $C \leq \log x/\sqrt{x}$, and that function is unimodal with maximum $2/e$ at $x = e^2$; solving the equality is a Lambert- W substitution.

$E_C(x) \leq (\log x)^2 \iff C \leq \varphi(x) := \log(x)/\sqrt{x}$ for $x > 1$. Now $\varphi'(x) = (2 - \log x)/(2x^{3/2})$, so φ increases on $(0, e^2)$, decreases on (e^2, ∞) , and attains its maximum $\varphi(e^2) = 2/e$ at $x = e^2 = 7.389056\dots$. Hence the superlevel set $\{x > 1 : \varphi(x) \geq C\}$ is non-empty iff $C \leq 2/e$ and, by unimodality with $\varphi \rightarrow 0$ at both ends, is a compact interval. Solving $C\sqrt{x} = \log x$ with $t = \log x$ gives $te^{-t/2} = C$, i.e. $(-t/2)e^{-t/2} = -C/2$, i.e. $-t/2 = W(-C/2)$, whence the stated endpoints; two real branches exist iff $-C/2 \geq -1/e$, i.e. $C \leq 2/e$. Finally the set is bounded, so it contains finitely many primes, and away from it $E_C(x) > (\log x)^2 > T_n$ off the exception set $\{1, \dots, 7, 10\}$. $\square$

Both published constants, $22/25 = 0.88$ and $21/25 = 0.84$, are *above* the critical $2/e$, so both give the empty set. Constants far below it buy a bounded initial segment and then stop: $C = 0.1$ covers $p \in (1.111, 8099)$, and $C = 0.01$ covers $p \in (1.010, 2\,122\,265)$. To certify F merely up to the *already verified* frontier one would need $C \leq (L^2 - L - 1)/(\sqrt{p}L) = 1.009 \cdot 10^{-8}$ at $p = 2^{64}$ — not a constant improvement but the abandonment of the $\sqrt{p}$ scale.

Corollary 7.9 (Even $\limsup = 0$ does not help; [P]). *Suppose $g_n = o(\sqrt{p_n} \log p_n)$. This certifies F at no index.*

Proof. Idea: $o(\sqrt{p} \log p)$ does not even imply $g_n < \log^2 p_n$, so it cannot imply $g_n < T_n$.

The hypothesis is compatible with gaps far above the threshold: a sequence with $g_n = \sqrt{p_n}L_n/\log \log p_n$ satisfies $g_n/(\sqrt{p_n}L_n) \rightarrow 0$ while $g_n/L_n^2 = \sqrt{p_n}/(L_n \log \log p_n) \rightarrow \infty$. Killing the constant does not change the scale, and by Theorem 7.7 the scale is the whole obstruction. $\square$

Caveat 7.10 (Provenance of the hypothesis of Corollary 7.9). That $\limsup = 0$ holds under RH together with Montgomery’s pair correlation conjecture is *reported*, not proved, by [CMS19, §1.2, after eq. (1.14)], who attribute it to three works not opened in this study. It is therefore second-hand and is attributed as such here, at a strictly weaker tier than the rest of that citation. Corollary 7.9 does not depend on the statement being true — it says that *even if* it holds, nothing is certified — so the weak tier does not propagate into the verdict.

7.4 The material implication $\text{RH} \Rightarrow F$

Theorems 7.3–7.8 refute the *route*. They say nothing about the *proposition*. Conflating the two is the failure mode this section most needs to avoid, so we state plainly: to refute $\text{RH} \Rightarrow F$ one must establish $\text{RH} \wedge \neg F$, i.e. prove the Riemann Hypothesis *and* exhibit a counterexample to F ; to prove it one must derive F from RH. Neither is available and no computation bears on either. **The implication is undecided here and we do not pretend otherwise.** What can be proved is a lower bound on the strength of what such a proof would yield.

Theorem 7.11 ([P]). *Suppose $\text{RH} \Rightarrow \text{F}$ were proved. Then, as an immediate corollary, one would have proved*

$$\text{RH} \implies (\forall k > 9 : g_k < L_k^2 - L_k - 1),$$

an RH-conditional Cramér-type prime-gap bound at $\log^2$ scale, uniform in k , with leading constant 1.

Proof. Idea: Compose with the unconditional necessary condition (2).

Compose the hypothesised implication with [Kou15a, §2, Thm. 1], which is the unconditional implication (2). $\square$

Corollary 7.12 (The size of the required advance; [P]). *The ratio of the best published RH-conditional bound to the bound Theorem 7.11 would deliver is*

$$\frac{\frac{22}{25}\sqrt{p_n} L_n}{L_n^2 - L_n - 1} \longrightarrow \infty, \quad \text{the ratio growing like } \frac{22}{25} \cdot \frac{\sqrt{p_n}}{L_n},$$

with value $8.72 \cdot 10^7$ at $p = 2^{64}$ and unbounded thereafter.

Caveat 7.13 (No difficulty ordering is claimed). Corollary 7.12 is a *distance between two statements*, not a measure of proof effort. Nothing here says such a proof must first pass through a sharpened $\sqrt{p}$ -scale bound, and nothing here orders $\text{RH} \Rightarrow \text{F}$ against any other open problem by difficulty. In particular this paper does *not* assert that F is harder than RH .

Corollary 7.14 (The band left for a candidate hypothesis; [P]). *Let H be any hypothesis of gap-bound shape with $H \Rightarrow \text{F}$. Composing with (2) gives $H \Rightarrow (\forall k > 9 : g_k < L_k^2 - L_k - 1)$. So every sufficient hypothesis is at least as strong as the necessary condition; and by (3) a hypothesis only an additive 0.17 stronger than that bound already suffices. Hence a candidate gap-bound hypothesis can sit only in the band between $L^2 - L - 1$ and $L^2 - L - 1.17$: outside it, a candidate is either too weak to give F or is itself a sufficient condition, i.e. F with a constant changed. **The band has width 0.17.***

Remark 7.15. Corollary 7.14 does *not* say that every sufficient H is within 0.17 of F — plainly false, since $g_k < L^2 - L - 5$ is sufficient and far stronger. The sandwich pins F , not the class of hypotheses that imply it. Corollary 7.14 uses (3), hence Axler (Caveat 5.5); the inequality *direction* is robust to the constant — any b with $1 < b < \infty$ gives the same conclusion with $b - 1$ in place of 0.17 — but the numeral 0.17 inherits that dependency. The corollary is also stated for hypotheses of gap-bound shape only.

7.5 Cramér’s hypothesis does not entail F over integer sequences

The last escape is: assume Cramér’s conjecture instead of RH . Write

$$(\text{Cr}) \quad \limsup_{n \rightarrow \infty} g_n / L_n^2 \leq 1. \tag{5}$$

Lemma 7.16 ([P]). *For every $\delta > 0$ there are infinitely many n with $g_n < (1 + \delta)L_n$.*

Proof. Idea: If all gaps were eventually $\geq (1 + \delta)\log p_n$, the primes would be sparser than the prime number theorem allows.

Suppose not: $g_n \geq (1 + \delta)L_n$ for all $n \geq N_0$. Summing, $p_N \geq (1 + \delta) \sum_{n=N_0}^{N-1} L_n$. By the prime number theorem $p_n \sim n \log n$, so $L_n \sim \log n$ and $\sum_{n \leq N} L_n \sim N \log N$. Hence $p_N \geq (1 + \delta + o(1))N \log N$, contradicting $p_N \sim N \log N$ for large N . $\square$

Theorem 7.17 (Counter-model; [P]). *Assume (Cr). Then there is a strictly increasing sequence $(q_n)_{n \geq 1}$ of positive integers with*

1. $q_n = p_n + O((\log n)^2 \log \log n)$;
2. $\limsup_{n \rightarrow \infty} (q_{n+1} - q_n) / (\log q_n)^2 = 1$, so q satisfies (Cr);
3. $q_{n+1}^{1/(n+1)} \geq q_n^{1/n}$ for infinitely many n , so q violates the Firoozbakht inequality infinitely often.

Hence (Cr) does not entail F in the theory of strictly increasing sequences of positive integers.

Proof. Idea: Take the primes and, at a sparse sequence of indices where the gap happens to be small, inflate that one gap up to exactly $\lceil \log^2 \rceil$; the inflations are so rare and so small that they do not disturb the limsup or the density, but each one breaks the Firoozbakht inequality.

By Lemma 7.16 with $\delta = 1$, for each $k \geq 1$ let $n_k := \min\{n \geq 2^{2^k} : g_n \leq 2L_n\}$, which exists, and $n_k \rightarrow \infty$. Define recursively

$$J_k := \lceil (\log q_{n_k})^2 \rceil - g_{n_k}, \quad q_n := p_n + \sum_{k: n_k < n} J_k;$$

the recursion is well founded because q_{n_k} depends only on $J_1, \dots, J_{k-1}$.

Positivity and monotonicity. $g_{n_k} \leq 2L_{n_k}$ while $\lceil (\log q_{n_k})^2 \rceil \geq L_{n_k}^2$, and $L^2 > 2L$ for $L > 2$, i.e. $p > e^2$. Start the construction at the least k_0 with $p_{n_{k_0}} \geq 11$ and discard $k < k_0$; this is legitimate because the conclusion concerns infinitely many indices, and it makes every $J_k \geq 0$. Then q is p plus a nondecreasing step function, hence strictly increasing.

Item 1. $q_n - p_n = \sum_{k: n_k < n} J_k$ with $J_k \leq \lceil (\log q_{n_k})^2 \rceil$. Since $n_k \geq 2^{2^k}$, the number of terms with $n_k < n$ is at most $\log_2 \log_2 n + O(1)$, and each $J_k \leq (\log q_n)^2 + 1 = O((\log n)^2)$. Hence $q_n - p_n = O((\log n)^2 \log \log n)$.

Item 2. At $n = n_k$, $q_{n_k+1} - q_{n_k} = g_{n_k} + J_k = \lceil (\log q_{n_k})^2 \rceil$ by construction, so the ratio is $\lceil (\log q_{n_k})^2 \rceil / (\log q_{n_k})^2 \rightarrow 1$. At $n \notin \{n_k\}$, $q_{n+1} - q_n = g_n$ and $\log q_n \geq \log p_n$, so the ratio is at most g_n / L_n^2 , whose limsup is ≤ 1 by (Cr). Hence the limsup is exactly 1.

Item 3. Write $T_n^{(q)} := q_n(q_n^{1/n} - 1)$; the derivation of Lemma 2.1 is purely formal and applies to any strictly increasing positive sequence, so the Firoozbakht inequality fails at n iff $q_{n+1} - q_n \geq T_n^{(q)}$. The index n is unchanged by the construction, and $q_n = p_n(1 + \varepsilon_n)$ with $\varepsilon_n = O((\log n)^2 \log \log n / p_n)$, which tends to 0 faster than any negative power of L_n . Since $\partial T_n / \partial p_n = O(L_n^2 / p_n)$, the perturbation moves the bar by $O(\varepsilon_n L_n^2) = o(1)$; with $T_n = L_n^2 - L_n - 1 + o(1)$ [Kou15a, §5, Thm. 5] this gives $T_n^{(q)} < (\log q_n)^2$ for large n . At $n = n_k$ the gap is $\lceil (\log q_{n_k})^2 \rceil \geq (\log q_{n_k})^2 > T_{n_k}^{(q)}$. So the inequality fails at every sufficiently large n_k . $\square$

Remark 7.18 (What Theorem 7.17 does and does not close). It closes the reading “Cramér’s limsup hypothesis entails F” as a counter-model result. It does *not* prove (Cr) $\not\Rightarrow$ F as a material implication about the primes: p is one fixed sequence, and if both (Cr) and F happen to be true the implication is vacuously true. Theorem 7.17 quantifies over sequences, on purpose. The stronger informal reading — “no derivation using only growth and distribution can succeed” — is a *gloss*, not a corollary: the drift in item 1 is small enough that no unconditional $\pi(x)$ estimate in our toolbox separates q from the primes (the induced displacement of the counting function is $O(\log x \log \log x)$, against a Dusart bracket of width $0.2762 x / \log^2 x$), so a derivation appealing only to (Cr), the prime number theorem and such estimates would have to prove F for q as well. That argument is only as good as its unformalised premise about which appeals a derivation makes, and the class it names is contingent on what this study fetched. **Quote the theorem, not the gloss.**

Structurally, Theorem 7.17 explains why the additive 0.17 of Corollary 7.14 is not a technicality: (Cr) controls a limsup, whereas F needs a uniform bound at *every* index with the second-order term pinned. The distance between “asymptotically at most 1” and “below $L^2 - L - 1$ always” is exactly the room the construction lives in.

7.6 Summary of the five readings

Reading of “an RH-conditional bound settles F”	Verdict
The sharpest published RH-conditional gap bound certifies F cofinitely	refuted — one index in range (Thm. 7.3)
Some bound $Cp_n^\theta(\log p_n)^A$, $\theta > 0$, certifies F beyond finitely many n	refuted (Thm. 7.7)
Some envelope $C\sqrt{p}\log p$, any $C > 0$, certifies F beyond finitely many n	refuted ; critical constant $2/e$ (Thm. 7.8)
Cramér’s $\limsup \leq 1$ entails F over integer sequences	refuted by counter-model (Thm. 7.17)
RH $\Rightarrow$ F as a material implication	undecided ; Thm. 7.11 bounds the required strength

8 The smooth model

It is worth recording why the difficulty is entirely in the fluctuation.

Proposition 8.1 ([P]). *The function $x \mapsto (x \log x)^{1/x}$ is strictly decreasing on $x \geq 5$.*

Proof. Idea: Take logarithms; the derivative of $\log(x \log x)/x$ has the sign of $1 + 1/\log x - \log(x \log x)$, which is negative once x is large enough, and $x = 5$ is large enough.

Write $\psi(x) := \log(x \log x)/x$. Then $x^2\psi'(x) = (1 + 1/\log x) - \log(x \log x)$. The bracket $1 + 1/\log x$ decreases and $\log(x \log x)$ increases, so ψ' has at most one sign change; evaluating, the bracket at $x = 5$ is 1.6213 while $\log(5 \log 5) = 2.0853$, so $\psi'(5) < 0$, and $\psi' < 0$ for all $x \geq 5$. Hence ψ , and therefore $(x \log x)^{1/x} = e^{\psi(x)}$, is strictly decreasing on $[5, \infty)$. $\square$

Since $p_n \sim n \log n$, Proposition 8.1 says that the *smooth model* of Firoozbakht’s conjecture is true and elementary. The entire difficulty of F localises in the fluctuation of p_n around $n \log n$. The safe stated range is $x \geq 5$, not $x \geq 4$: the relevant bracket changes sign just above 4 (+0.0084 at $x = 4$, −0.0193 at $x = 4.05$). Proposition 8.1 was *not* formalised in Lean in either round (§3), and it remains the highest-leverage formalisation target the work has: it is the only node in the formalisation plan that is a genuine theorem rather than a definition or a finite check, and it is fully within Mathlib’s reach.

9 Computational evidence

9.1 Design

Two independent legs, with independent code paths, swept the primes exhaustively to 10^{11} — 4 118 054 812 consecutive prime pairs, no pruning, no sampling, no assumed lemma. The protocol is two-tier and the tiering is the point:

- **Screen.** float64 over all pairs, on $\rho_n = g_n/T_n$. This statistic has $O(1)$ operands and so does not carry the catastrophic cancellation that afflicts the ratio statistic of §9.5.
- **Escalation.** Every pair with $\rho_n \geq 0.90$ is re-decided in arbitrary precision with an explicit error budget, and the certified verdict function *raises* rather than returns when the margin lands inside that budget.
- **Calibration.** The high-precision path is checked against the exact integer criterion $p_{n+1}^n < p_n^{n+1}$ for every $n \leq 5000$ in one leg and $n \leq 4000$ in the other. Zero disagreements.

The escalation discipline deserves emphasis because it inverts the usual default: *the silent failure here is in the verification direction*. A probe that only breaks on a detected violation reports “no counterexample” out of numerical noise. This one cannot. Round 2 applied the same discipline with the sign flipped, in the one place where it mattered: the refutation of (P6′-pair) (Theorem 4.3) is the outcome that is bad news for the attack, so its witnesses were recomputed at 60 digits even though they stand 10^8 – 10^{12} ulps clear.

Round 2 added two further sweeps, written from the statements rather than from any round-1 code path: one to 10^9 (50 847 533 indices) and one to $2 \cdot 10^8$ (11 078 936 indices). Both reproduce round 1’s headline statistics to every digit quoted. The smaller of the two is explicit that the 10^9 decade is one beyond its own sieve and is therefore *not* independently confirmed by it, and that nothing in its report depends on that decade; that is the discipline worth carrying.

9.2 Results

Quantity	Value
consecutive prime pairs checked ($p_n < 10^{11}$)	4 118 054 812
violations of F	0
$\max_{n \geq 10} \rho_n$	0.8317570 at $p_n = 25\,056\,082\,087$, $g = 456$
$\max_{n \geq 10} g_n/L_n^2$	0.7953487, same prime
largest gap in range	464 at $p = 42\,652\,618\,343$
maximal-gap records in range	40
sieve validation	$\pi(10^9)$, $\pi(10^{11})$ match standard values
(P6′-gov), (P6′-min), (P6′-rec), $T_{\mu(n)} \leq T_{r(n)}$ below 10^9	0 exceptions each
record steps checked for (P6′-rec) below 10^9	29
(P6′-pair) exception indices below 10^9	17 , in two clusters (Prop. 4.5)

Only two pairs in the whole range reach $\rho \geq 0.9$, and both are at $n < 5$ ($\rho_2 = 0.9107$, $\rho_4 = 0.9120$), which is why the record is quoted for $n \geq 10$.

9.3 The oscillation of T , and a three-way disputed denominator

The fact that motivates §5 is that T is not monotone. Quantified:

range	$T_{n+1} < T_n$, $n \geq 10$	$T_{n+1} < T_n$, all n
$3 \cdot 10^6$	121 238/216 806 = 55.9200%	121 239/216 815 = 55.9182%
10^9	—	$\approx 57.88\%$

Caveat 9.1 (The figure is range-dependent, and its denominator was disputed three times). Two riders travel with this figure and neither may be dropped. First, it is **range-dependent** — 55.92% at $3 \cdot 10^6$ against 57.88% at 10^9 — and must never be quoted without its bound and its convention. Second, the denominator has a history worth recording, because it is a case where the review process was wrong twice in the same direction. Every *numerator* above agrees across every document and both rounds. The *denominators* were reported one lower (216 805 and 216 814) by one round-2 leg, and a round-1 review had called the higher figure an off-by-one error while a round-2 leg re-affirmed that call. Three mutually independent recomputations — including one written from the statements with its own sieve at 60 digits — agree on $\pi(3 \cdot 10^6) = 216\,816$ and hence on 216 815 consecutive steps in total and 216 806 with $n \geq 10$. The lower figures are consistent with a T -array truncated to the gap-array length before differencing, i.e. an implementation artefact rather than a counting convention. **We carry** 121 238/216 806 **and** 121 239/216 815, and record that the concept card downstream

legs read first still carries the disputed lower denominator, so the discrepancy is live in the corpus (§11.3).

Whichever convention is used, the statistic is **not diagnostic** for any of the four predicates of Definition 4.1: it measures single steps, while every predicate compares an index to a governor many steps back.

9.4 Verification of the lemmas of §5–§6

The following items are cited by number from the proofs above. All were recomputed independently of the code paths that produced the sweep.

- V1 $\max\{g_k : p_k < 60\,184\} = 72$, attained at $p = 31\,397$, by exact integer arithmetic over the 6 076 gaps below 60 184; and $B(60\,184) = 109.008\dots > 72$. Also: $p_{n+1}^n < p_n^{n+1}$ for every one of those 6 076 indices in exact integer arithmetic — this is hypothesis (H1) of Theorem 6.4, discharged.
- V2 $S(29) = 6.8013853766\dots > 6 = \max\{g_j : j \leq 9\}$ — hypothesis (ii) of Theorem 5.6.
- V3 Lemma 6.2 holds at every prime in $[60\,184, 10^{11}]$; the tightest slack is $T_n - B(p_n) = +0.0798914728\dots$, attained at the very bottom of the range ($p = 155\,893$).
- V4 Lemma 6.1 holds at every prime in $[60\,184, 2 \cdot 10^7]$; zero failures. The three Axler bounds of Caveat 5.5 were likewise tested at $x = p_n$ over the primes below 10^8 , each within its own stated range: zero failures each.
- V5 $B(2^{64}) = 1919.13798349753288\dots$ (Proposition 6.6), and the largest even g with $S^\sharp(g) \leq 2^{64}$ is 1918.
- V6 The sign change of $h(p) = p - 25(\log p)^3(\log p - 1.1)$ is at $p^* = 777\,600.744\dots$ (Proposition 6.8).
- V7 Table lookup for the whole in-run range: of the 209 gap sizes occurring below 10^{11} , 55 are excluded outright by $S^\sharp(g) < 60\,184$ and 154 need the table test; **none is unsettled**. The minimum safety factor $q(g)/S^\sharp(g)$ is 5.337 at $g = 112$, the maximum 203.9 at $g = 334$.
- V8 At the published record Cramér–Shanks–Granville point $p = 1\,693\,182\,318\,746\,371$, $g = 1132$ [Sloa]: $B(p) = 1191.009$, so the record gap is safe with a margin of 4.955%.
- V9 The certificate that scales: splitting $[60\,184, 10^{11}]$ into 22 geometric windows $[a, b)$ and checking $\max\{g_n : p_n < b\} < B(a)$ certifies every window, consuming *only* the 40 maximal-gap records — no sieve, no $\pi(x)$ computation, no per-prime check. This is the shape in which a frontier like 2^{64} is actually certified.
- V10 The two small-branch inputs of §6.5: the largest prime gap below 10^8 is 220, attained at $p = 47\,326\,693$; the largest below 6 690 557 is 154, attained at $p = 4\,652\,353$. Both by exact integer arithmetic over an in-run sieve.
- V11 The constants of §6.5, all recomputed at 40–60 digits from the statements: $e^{-0.0516} = 0.949708674346\dots$; $e^{-0.0017569} = 0.998244642445\dots$; $e^{-0.0043636} = 0.995645906670\dots$; the majorant maxima 0.051599027 and 0.0017568759; the exact (non-majorised) requirements 0.051493457 and 0.0017560603; and the round-1 defect of Caveat 6.13: required separation 0.169339812744... under the printed lemma against 0.004363567696... under its tight form, ratio 38.8137468954...

V12 The barrier of Theorem 3.1, checked numerically as a sanity companion to the kernel proof: $p_n^{1+1/n} < 2p_n$ at every $2 \leq n \leq 2 \cdot 10^4$ and $p_n < 2^n$ at every $2 \leq n \leq 216\,816$ in exact integer arithmetic, zero failures each; and Bertrand does *not* certify at $n = 1$, where $2^2 = 4 = 2 \cdot 2$ — the one index the theorem excludes.

V13 $\log 2^{64} = 44.36141955583649\dots$, and the RH critical constant $\max_{x>1} \log x/\sqrt{x} = 0.7357588823428846\dots = 2/e$ at $x = e^2$, to 60 digits (Theorem 7.8).

Remark 9.2 (Stability of the table test across four decades). The minimum safety factor $q(g)/S^\sharp(g)$ is 5.3371 at $g = 112$ at every one of $X = 10^7, 10^8, 10^9, 10^{10}, 10^{11}$. Four decades of new data and 134 new gap sizes produced nothing tighter than a case already visible in the first decade ($g = 112$, first occurring at $p = 370\,261$). The table test neither tightens nor loosens as the range extends.

9.5 Three refutations of claims about the evidence

Computation refutes nothing about F. It did refute three claims about the evidence, and recording them is the point of this subsection.

(1) The “0.9999984 near-miss” is an artefact. The statistic $\max_n n \log p_{n+1}/((n+1) \log p_n)$ reads 0.9999984 below $3 \cdot 10^6$ and has been read as evidence that F nearly fails. It is not. In a synthetic universe where every gap equals 2 (so that F holds by a mile) the same statistic reads 0.99999991 at $n = 10^7$, while ρ_n there is 0.059 — an order of magnitude below the bar. The statistic measures $1/n$. Only ρ is diagnostic:

X	10^7	10^8	10^9	10^{10}	10^{11}
max ratio statistic	0.99999937	0.99999994	0.999999993	0.9999999992	0.99999999992
$\max_{n \geq 10} \rho_n$	0.7605	0.7896	0.7896	0.7896	0.8318

(2) “The tightest cases sit at record gaps” does not survive the range. The observation that the tightest ρ cases occur at record gaps was a print truncation of the first six entries. At 10^{11} : 22 of the top 40, and the fourth-tightest case among four billion pairs is *not* at a record index ($g = 444$ at $p = 36\,172\,730\,063$, $\rho = 0.78502$). Read precisely: if the pruning rule holds, a record-index search still misses no *counterexample*; what it misses is *near-misses*. So the observation cannot be used as evidence for the pruning rule that produced it.

(3) More sieve is not more evidence. Two decades of extra sieving ($10^8 \rightarrow 10^{10}$) produced no new near-miss: the record ρ sat at 0.7896 ($g = 210$, $p = 20\,831\,323$) throughout. It moved exactly once, at 10^{11} , and it moved *at the next maximal gap*. Sieving buys the next maximal gap and nothing in between. The record that would matter — $\rho \approx 0.948$ at $p \approx 1.693 \cdot 10^{15}$ [Sloa] — is 4.2 decades above anything reached here. The compute that would matter goes into gap *tables*, not prime *counts*.

9.6 A counter-model for first-failure maximality

Proposition 9.3 ([C]). *There is an explicit strictly increasing sequence of positive integers whose first Firoozbakht failure occurs at a non-record gap.*

Proof. Idea: Take the real primes up to 1327, then a large jump, then a long dense run of gaps of 2, then a jump smaller than the first one; the dense run drives the index up so far that the later, smaller gap is the one that breaks the inequality.

Take the primes up to 1327, then a jump of 40, then 100 jumps of 2, then a jump of 38. The first Firoozbakht failure of this sequence is at index 323 on the gap of 38, which is not a record (the earlier jump of 40 is larger). Verified in exact integer arithmetic, so rounding cannot be blamed. $\square$

The consequence is sharp: **any proof of first-failure maximality that does not consume an arithmetic density input is wrong**, because the statement is false for sequences satisfying everything except primality. Note that Theorem 4.11 respects this: it consumes a hypothesis about T that is a statement about the primes. The good news is that a weak input suffices for this mechanism: the dense run the counter-model needs exceeds the unconditional Montgomery–Vaughan cap $2y/\log y$ by a factor that *grows* with scale — 1.41 at $p \approx 10^3$, 4.18 at $4 \cdot 10^8$, 9.13 at 10^{18} — so the counter-model is excluded unconditionally, and increasingly comfortably. (The Montgomery–Vaughan bound is not covered by a source-ledger row in this work; cf. Caveat 6.20.)

A second, independent route is also closed: bounding T_m above and T_n below with explicit $\pi(x)$ estimates *at the two endpoints separately* demands record gaps of size at least $5.3 \cdot 10^4$ at $p = 10^6$ (Dusart) or $3.7 \cdot 10^3$ (Axler), against actual records of size $\approx \log^2 p \approx 176$; and past $\approx 10^{10}$ (Dusart) the criterion is unsatisfiable at *any* gap size. The bound spread dwarfs the quantity being resolved, and no sharpening of constants rescues the route. This is not a contradiction of §6.5: there the two endpoints are separated by a fixed multiplicative factor, which is exactly the regime the two-sided route cannot reach.

10 The tension that will not resolve itself

The strongest reason to believe F is *false* is not a computation and not a theorem. Cramér’s model, in Granville’s corrected form, predicts $\limsup g_n/\log^2 p_n \geq 2e^{-\gamma} \approx 1.1229 > 1$ [Gra95, preprint p. 12], which is incompatible with F via (2). This is [H]: a heuristic, not a test. Granville’s own sign is $\gtrsim$ — “suggests”, not “proves”.

Symmetrically, the strongest reasons to believe F is *true* are the verified range (Theorem 2.3) and the fact that the record Cramér–Shanks–Granville ratio 0.9206, standing since 1999 [Sloa], sits comfortably below the Firoozbakht-implied ceiling of $1 - 1/\log p \approx 0.9715$ at that size — about 5% of headroom. Neither is a proof and neither is close to one.

Caveat 10.1 (Cramér did not conjecture $\limsup = 1$ about the primes). He *proved* it about his urn model, with probability 1 [Cra36, p. 27], and *suggested* the analogue for the primes [Cra36, p. 24, eq. (4)]. Granville frames it the same way. Any sentence of the form “Cramér conjectured that $\limsup g_n/\log^2 p_n = 1$ ” is a compression the source does not literally support.

11 What remains open, and what this paper does not establish

This section is long on purpose. A paper reporting an attack that failed to settle its target is exactly the paper that gets over-quoted downstream.

11.1 F itself, and why it is not close

The load-bearing obstruction is unchanged by anything above: **any proof of F yields $g_n = O(\log^2 p_n)$ unconditionally**, via (2). The best known unconditional gap bound is $g_n = O(p_n^{0.525})$ [BHP01]; under RH it improves only to $\approx \sqrt{p_n} \log p_n$ [CMS19]. Both are powers of p_n ; F needs polylogarithmic. Any proposal of a “direct proof” must say how it clears this, or it is proposing something already beyond the field.

Round 2 sharpened this in three independent places, produced by three legs that were not talking to each other — which is worth recording, because agreement reached that way is evidence and agreement reached by copying is not:

- **Formally**, the only prime-gap input Mathlib carries is proved insufficient at every index (Theorem 3.1, [K]).
- **Analytically**, every envelope $Cp^\theta(\log p)^A$ with $\theta > 0$ fails beyond finitely many n , because the bar sits at $\theta = 0$ (Theorem 7.7); and any hypothesis sufficient for F must itself deliver the full $\log^2$ -scale uniform bound with leading constant 1 and second-order term pinned, while a hypothesis only 0.17 stronger already suffices — the band left for a candidate gap bound has width 0.17 (Corollary 7.14).
- **Numerically**, the crossover table of §7: the available slack overtakes what is needed from $n = 245$ on unconditionally and from $n = 3$ on under RH.

Compounding it: **there is no induction mechanism.** g_n is not constrained by $g_1, \dots, g_{n-1}$; knowing F up to n gives no leverage at $n + 1$. Any proposed inductive proof must first supply the missing mechanism.

On the refutation side, the best large-gap results [For+16] reach an iterated-log factor above $\log n$ but remain a full power of $\log$ below what a counterexample needs. Explicit constructions fail for a second, independent reason: they place the gap at an *unspecified* location, while F needs the gap and the count $\pi(p_n) = n$ at the *same* point. Correspondingly, although $\neg F$ is Σ_1 and finitely certifiable in principle, the certificate must certify the *rank* n , not merely the two primes.

11.2 The most tractable open node

It is Proposition 6.19: an unconditional upper bound on the number of primes in a short interval, sharp to a factor $1 + 2/\log p$, where Brun–Titchmarsh gives about 2. It is stated in closed form, it is the exact obstruction inside the residual window of Theorems 6.14 and 6.15, and it is an open problem in analytic number theory rather than a lookup. Two further nodes worth naming: proving (P6'-rec), or measuring it as a first-class obligation (Proposition 4.7 needs it and nobody was measuring it under its own name until round 2); and formalising Proposition 8.1.

11.3 Named defects and unreconciled seams in the underlying corpus

The body of work this paper draws on was subjected to adversarial review in both rounds. Round 1's two blocking defects are repaired (§1.4). Round 2's review returned **three** new blocking findings, and this is the honest reading of that: the count went up, and the species changed. *None of the three is a mathematical error, and none touches F.* All three are seams between artifacts that no leg owned.

1. **Two correct, incompatible repairs of the same defect.** Two legs independently repaired the round-1 bound of Caveat 6.13, correctly, into two different theorems with two different constants, off two different rows of the same source table, with neither citing the other. Both were verified correct. The corpus contains no rule for choosing. This paper chooses — Theorem 6.15, on the ground that its source row exists in both editions — and states that it is choosing (Remark 6.17). **Neither upstream document has been amended.**

2. **Contradictory source tiers for the same source.** One leg records [Axl14] as opened and read at the locator; another still labels it unopened at every use and attaches to its own theorem the instruction that it must never appear in a sentence containing the word “unconditional”. That instruction is now too strict on *tier* grounds and exactly right on *row* grounds (Caveat 5.5(b)). A reader of the second document alone gets the wrong tier. This paper is written in the corrected tier throughout; the second document has not been amended.
3. **An edition-fragile citation.** The table row underlying the retired repair exists in the preprint edition only, and the artifact that *depends* on it does not carry the flag, though the concept card describing the source does. This paper resolves its own exposure by not carrying that theorem (Remark 6.17); the flag still has to be written into the upstream document if it is kept.

Two further live defects sit *inside* the repairs, both labelling rather than arithmetic, and both already stated at their point of use: the “census of pairs” that counts indices (Caveat 4.6), and the “unconditional” label on Theorem 6.14 that rests on a verification height and an in-run sieve (Caveat 6.18). And three small numeric defects were found by recomputation, none touching any conclusion: an upstream document prints $e^{-0.0017569} = 0.99824467$ where the value is $0.9982446424\dots$; and the concept card that downstream legs read first carries the disputed denominator 216 805 (Caveat 9.1) together with a transposed sweep size, 50 847 503 for 50 847 533.

The pre-publication evidence gate on the underlying corpus stands at BLOCKED, with the adversarial-review leg named as the failing one, on the three findings above. The formal (kernel) leg passes outright — and *passing the kernel leg is not the claim “proved”*; the kernel verdict on F itself is that it was not provable within the budget. The adversarial-corpus leg passes (27/27). Nothing in §§2–10 rests on an unrepaired defect: no constant here comes from a broken derivation, the withdrawn ordering of Remark 4.9 is nowhere used, and the retired constant of Remark 6.17 appears only as the statement being retired. **This paper is an honest report on a blocked corpus. It is not a seal.**

Remark 11.1 (The structural reading, which is what two rounds bought). Round 1’s review named this failure mode and predicted its recurrence: a fan-out with no reconciliation stage, in which nobody owns the seams. Round 2 *widened* the fan-out, still supplied no reconciliation stage, and reproduced the prediction on itself. Every mathematical target round 2 was pointed at, it hit; what is missing is not mathematics but a single reconciliation pass. The useful conclusion — which one round could not have supplied, having only one measurement — is that more rounds *of this shape* will not converge. We state it here because a reader deciding whether to extend this work should know that the next increment is editorial, not mathematical.

11.4 Source provenance, and the citation audit that has not run

Every citation in this paper traces to a row of a source ledger built from scratch for this study: **22 rows** — **13 at L0** (primary source fetched, statement read at the recorded locator), **4 at L1** (fetched, but at an edition or pagination that could not be independently confirmed), **3 at L2_strong** (attested at a specific locator by two or more independent fetched L0 sources), **2 at L2_weak** (one such source), and **none at L3** (recall). No citation in this paper rests on recall.

Two of those rows are new, and how they got there must travel with them. [CMS19] and [Vis18] were fetched, version-pinned by digest and read at the locator by one leg of round 1, *proposed* as ledger additions, and never folded in; the round-1 citation audit of this paper’s predecessor consequently returned **BLOCKED**, naming exactly those two keys as tracing to no ledger row. The proposal has now been folded, verbatim as proposed, by the leg that authored this paper — which is not the leg that owns the ledger. What that leg independently verified is the bibliographic envelope of both sources (title, authors, journal, volume, pages, DOI, and the existence of the pinned preprint version, checked against the publishers’ listings); what it did *not* do is re-open either document at the interior locators. **Both rows carry a standing re-audit obligation.** We record this rather than presenting a clean ledger, because a leg minting rows outside its own remit is the same species of seam as §11.3’s three, made by this paper’s own author.

The citation audit for this paper has not been run, and no citation clearance is claimed. The round-1 audit of the round-1 paper returned **BLOCKED** and the downstream editorial gate consequently returned **REWRITE**; no round-2 audit exists. The load-bearing unopened or under-located sources are, in priority order:

1. [Gra95] — fetched and read, but at *preprint* pagination (pp. 1–16) against the journal’s pp. 12–28. Every locator must be re-expressed against the journal copy. This is the load-bearing citation of the entire refutation-side argument (§10), and it is now **priority 1**, Axler having been opened.
2. The published 2^{64} verification height itself, which this work reaches through a concept card resting on [OHP14] at the weakest ledger tier. [OHP14] was *not opened* (the publisher returned HTTP 403). It is the first-occurrence gap table on which Theorem 2.3 rests, and therefore the single largest external dependency of every statement about 2^{64} here — including Proposition 6.6 and the hypotheses of Theorems 6.14 and 6.15. If that table is incomplete, those statements say nothing. **Priority 2.**
3. The two new rows above. **Priority 3**, and the reason is the fold, not the sources.

Further, [Rib04] (the origin locator, p. 185) was not opened and rests on two independent citers; [Sha64] and [FJ17] were not opened and are cited for attribution only, their content being available at first hand elsewhere in the bibliography; and [BHP01] and [For+16] were read at abstract level only — adequate for the qualitative use both are put to, inadequate if any internal constant were quoted, which it is not. Caveat 6.20 names two standard results used only negatively and covered by no row at all.

11.5 Scale, stated plainly

The in-run exhaustive sweep reached $p < 10^{11}$, which is 8.27 decades short of 2^{64} . Several of the lemma verifications were run at the smaller scale $3 \cdot 10^6$, which is 12.8 decades short of 2^{64} . **Both sweeps verify the lemmas, not the range.** They must never be cited as a verification of F. The distance between what this study owns and the published frontier is exactly one external table and one inequality to check against it: for every even g occurring as a prime gap in $[10^{11}, 2^{64}]$, is $q(g) > S^\#(g)$? The checker is written and validated on data we own; supply the table and the frontier is re-derived in seconds.

Finally: “unconditional” means “no unproved hypothesis”, not “no computation”. Proposition 6.9 proves the computation cannot be removed. Any sentence of the form “F is unconditionally true below X ” must, to be honest, carry the table with it.

11.6 Dead ends, recorded so they are not rediscovered

- **Littlewood oscillation is irrelevant** to the threshold, its contribution being $O(L^2 \log \log \log x / \sqrt{x})$, which tends to zero.
- **Bertrand’s postulate is not merely useless here, it is provably closed** at every $n \geq 2$ (Theorem 3.1). There is no sharper version of Bertrand that helps, because the shortfall is a whole scale.
- **(P6’-pair) is false** (Theorem 4.3) and $(P6’\text{-gov}) \Rightarrow (P6’\text{-min})$ is invalid (Proposition 4.8). Do not attempt either.
- **The two-sided π -bound route** to the maximal-gap reduction cannot work (§9.6).
- **More sieving is not more evidence** (§9.5(3)). The record that would matter is 4.2 decades above anything reached here.
- **The table-free route** is permanently dead above $7.776 \cdot 10^5$ (Proposition 6.9).

12 Conclusion

Firoozbakht’s conjecture was neither proved nor refuted here, in either round. What was produced instead is: a machine-checked reduction of it to a prime-gap inequality, now joined by a machine-checked proof that the only prime-gap input the formal substrate carries is insufficient at every index; an unconditional, table-driven verification architecture that reproduces the published 2^{64} frontier from first principles and rests on a single explicit estimate for $\pi(x)$, with its near-record companion repaired and tightened; an exhaustive independent sweep to 10^{11} finding no counterexample and no near-miss, reproduced twice over; five theorems closing the Riemann-Hypothesis route as a route, with the quantifier that the first round dropped restored; and the outright refutation of the pruning lemma the attack had most quoted, together with a proof that the pruning never needed it — offset by three named, unrepaired seams in the underlying corpus and by a citation audit that has not run.

The most tractable genuinely open analytic node isolated here is the residual window of §6.5: an unconditional short-interval prime count sharp to a factor $1 + 2/\log p$, where Brun–Titchmarsh gives only a factor 2 (Proposition 6.19). It is stated in closed form. It is also still an open problem in analytic number theory, not a lookup.

The standing instruction with which this work was carried out is the one it ends on. *The conjecture is open. Do not write “Firoozbakht is true”. Do not write “Firoozbakht is false”. The Cramér–Granville tension is evidence about which way to bet, and nothing more.*

Acknowledgements and provenance

This paper was produced by **Noogram** as the editorial artifact of a structured, multi-leg attack conducted over two rounds on 25–26 July 2026. **It is the round-2 state of that work and supersedes, in place, the round-1 paper of the same name**; where the two differ, this one is current, and §1.4 states what changed and what did not. The legs contributing results reported here are, in round 1: a decomposition leg; a source-ledger leg; a concept-card leg; three proof-attempt legs; three computational-notebook legs; Lean skeleton and probe legs; an adversarial red-team corpus leg; an adversarial review leg; an evidence gate; and a synthesis leg. In round 2: three proof-attempt legs, a Lean probe leg (which contributed Theorem 3.1), an adversarial review leg, an evidence gate and a superseding synthesis leg.

The delivery posture of this artifact is **staged**: it reports a corpus whose evidence gate stands at BLOCKED (§11.3) and for which no citation clearance exists (§11.4). It is not a seal, and it should not be cited as though a clean gate stood behind it.

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